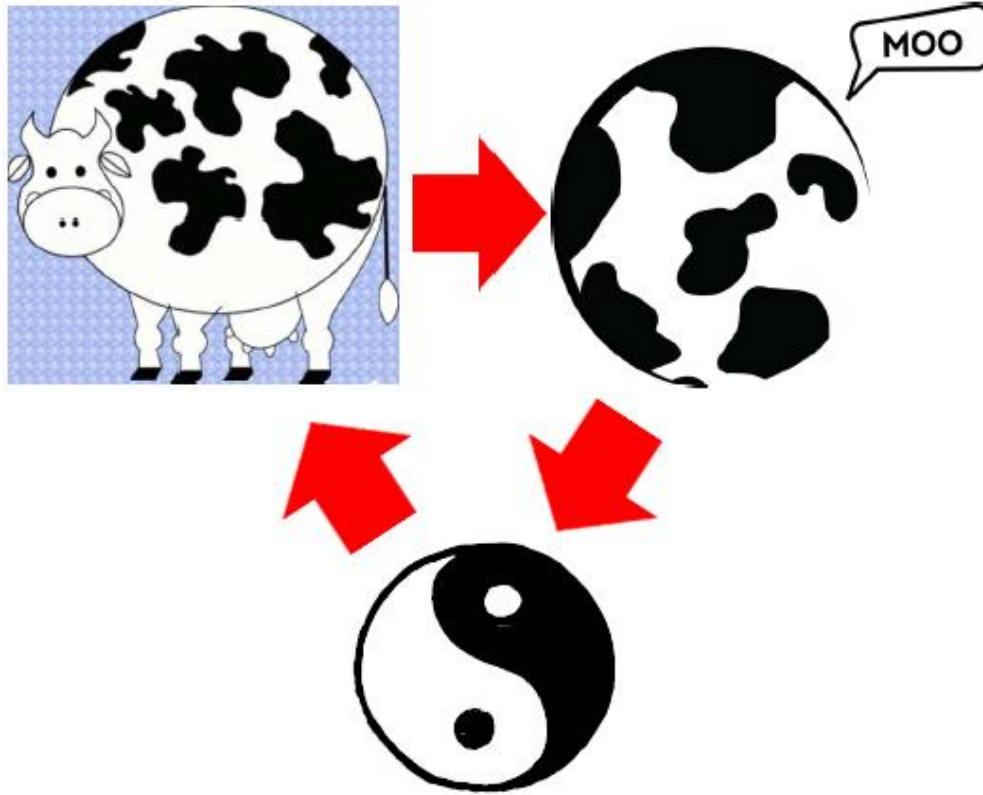


Complex Physics (2025)

Thematic introduction:

“More is different”



Thematic introduction:

“More is different”

Large systems=many degrees of freedom

Equilibrium $\leftrightarrow$ Non equilibrium

Organized by Energy $\leftrightarrow$ by Time

Critical point - Emergence...

→ long range correlations=power laws/fractals
(that is from $\exp(-x)$ to $1/x$)

Progression

- 1: Equilibrium & detailed balance & Ising model
- 2: Critical points (equilibrium)
- 3: Scaling, percolation, critical exponents, fractals, begin to introduce time (directed percolation)
-d/dt physics in discrete space:
 - 4: Random walks, Critical branching & Self organized criticality
 - 5: Stochastic simulations and Agent based models. Excitable media, Non equilibrium Ising like model & epidemic model.
 - 6: Networks & critical spreading on networks & scale free networks & pattern in networks & construction and emergence of scale free networks
 - 6X: Hypercycles, Origin of Life Modeling by Kuni Kaneko
 - 7: Econophysics: random walks in log space, Hurst exponents, aspects of bet hedging. Models to think about stock price fluctuations.
- overall ideas in self organization:
iterate a simple process many many many times.... Time as organizer!

Broad spectra of topics

- Connected *in part* by history (of invention 1924-1980 (Ising model) → 1980 (chapter 3) → 1990 (chapter 4) → 2000-2020 (chapter 5 & 6 & 7))
- Connected *in part* by power laws
- *In part* by universality... one rule to rule them all.... Here we will see some power laws that are recurrent, $1/x^{1.5}$ and $1/x^2$.
- universal methodology from physics to evolution, to social network to diseases, to economic systems
- From equilibrium towards various forms of self organization (using time)

Entropy:

$$S = -k_B \langle \ln(p_i) \rangle = -k_B \sum_{i \in \Omega} p_i \ln(p_i)$$

If S were measured in units of $\log_2(p_i)$ ($S \rightarrow I = S/k_B/\ln(2)$) it would count the *average* number of yes-no questions to locate a particular i :

For example, consider a 3-state system with $p_1 = 0.5$ and $p_2 = 0.25$ and $p_3 = 0.25$, then asking first whether or not the system is in state 1 will give full localization using 1 question with probability 1/2. In case it was not in state 1, one needs to use one more question to locate it. Thus one ask if it is in state 2, and would have fully localized it whether or not this is the case. Thereby one needs 2 questions with probability 1/2 also, giving an average number of questions of 1.5.

One particle is randomly placed
in below system

1/2	1/4
	1/4

Locate which box
the particle is in by
asking yes-no
questions

$$I = 0.5 * \log_2(4) + 0.5 * \log_2(2) \\ = 1.5 * \log_2(2)$$

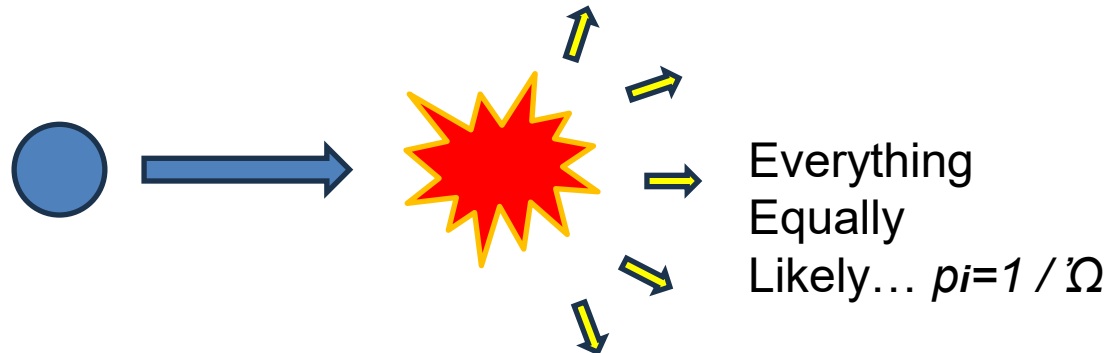
Equilibrium in an isolated system

All states in phase space Ω equally likely: $p_i = 1/\Omega$ and:

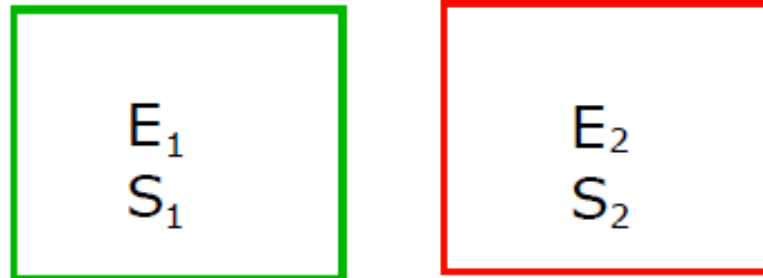
$$S = k_B \ln(\Omega)$$

... this is obtained after long time,

... or in practice after a lot of mess → everything equally likely
it's a good
minimal a-priori information
assumption

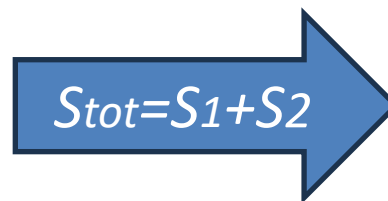


Thermally coupled systems



$$S(\text{tot}) = S_1 + S_2$$
$$E(\text{tot}) = E_1 + E_2$$

From independence: $\Omega(E) = \Omega_1(E_1) \cdot \Omega_2(E_2)$



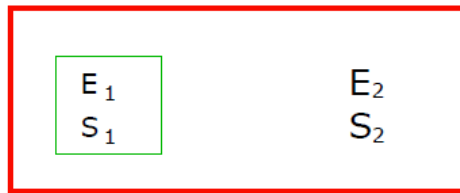
$$\frac{\partial S_{\text{tot}}}{\partial E_1} = \frac{\partial S_1}{\partial E_1} - \frac{\partial S_2}{\partial E_2} = 0,$$

Varying energy E_1 while fixing $E(\text{tot})$:

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2} \equiv \frac{1}{T},$$

This defines temperature.

Boltzmann distribution



$$S(\text{tot}) = S_1 + S_2$$

$$E(\text{tot}) = E_1 + E_2 \quad E_1 \ll E_2$$

$$\begin{aligned}
 p(E_1) &= \frac{\Omega_1(E_1)\Omega_2(E - E_1)}{\sum \Omega_1(E_1)\Omega_2(E - E_1)} \\
 \Omega_2(E - E_1) &= \exp\left(\frac{S_2(E - E_1)}{k_B}\right) \\
 S_2(E - E_1) &= S_2(E) - E_1 \frac{\partial S_2}{\partial E} \quad \text{and} \quad \frac{1}{T} = \frac{\partial S_2}{\partial E} \\
 p(E_1) &= \text{const} \cdot \Omega_1(E_1) \cdot \exp\left(-\frac{E_1}{k_B T}\right) = \text{const} \cdot \exp\left(\frac{-E_1 + T \cdot S_1}{k_B T}\right)
 \end{aligned}$$

If only one state

per energy level E_1 : $p(E_1) \propto \exp(-E_1/k_B T)$

Interpretation:

$$\begin{aligned} e^{-E/k_B T} &\propto 0.37 && \text{for } E \sim 1k_B T \\ &\propto 0.14 && \text{for } E \sim 2k_B T \\ &\propto 0.00005 && \text{for } E \sim 10k_B T \end{aligned}$$

$$P(E) = \exp(-E/k_B T) = \exp(-1) \exp(-1) \exp(-1)$$

In total $E/k_B T$ times,

It's multiplicative

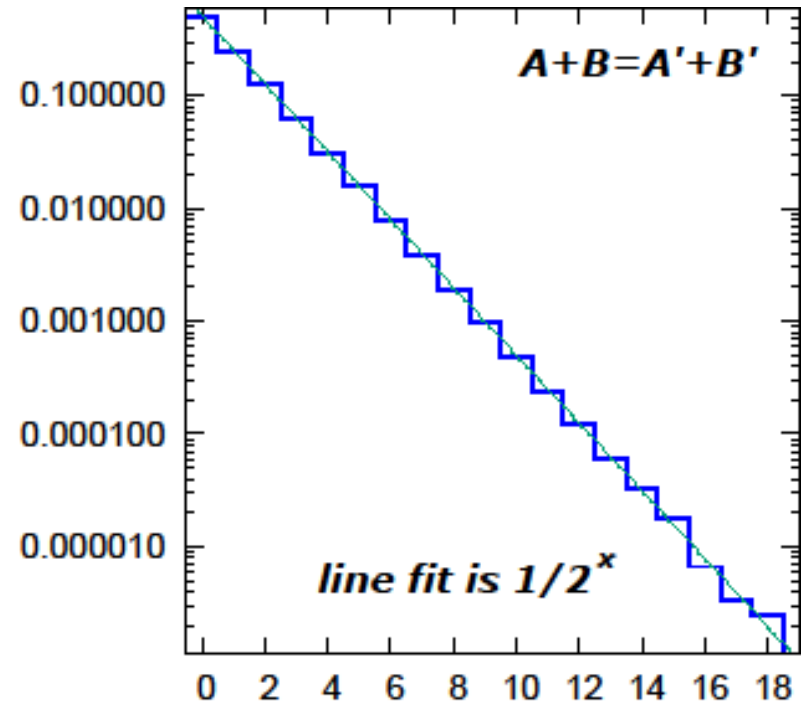
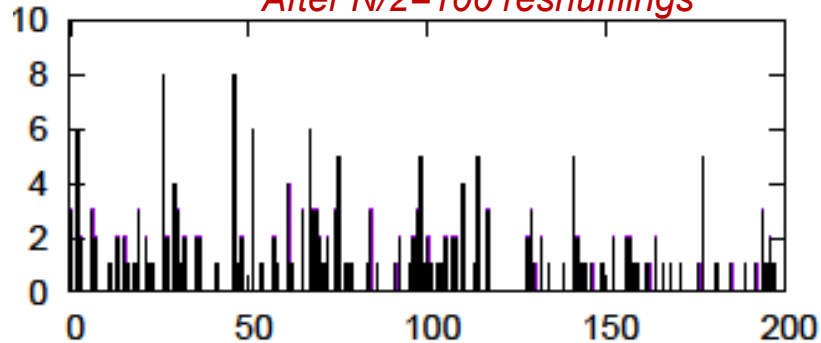
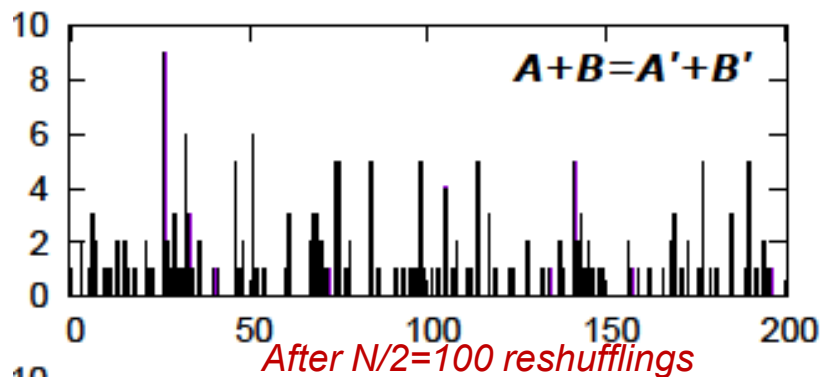
Because each term represents the <1 probability that in the next collision, then one more energy unit is added to our degree of freedom, instead of one energy unit being lost.

In equilibrium, the energy is narrowly distributed!

Illustration: Boltzmann distribution from random dynamics of Energy packages:

Initially all numbers = 1:

**Iterate: Take two random numbers (A,B) and randomly reshuffle
their sum (A+B) between them (giving A'+B')...
(idealized collision between particles exchanging energy)**



Partition Function:

=statistical weight, sum over probabilities

$$Z \equiv \sum_i \exp(-\beta \epsilon_i)$$

Where sum runs over all microstates i of the system

.....

When some states have the same energy $\rightarrow$

Counted together with prefactor = exp of their combined entropy.

$$\begin{aligned} Z &= \sum_i \exp(S(E_i)/k_B) \cdot \exp(-E_i/k_B T) \\ &= \sum_i \exp(-\beta(E_i - TS(E_i))) \end{aligned}$$

Partition sum Z

- Used in calculating averages:

$$\langle E \rangle = \frac{\sum E_i \exp(-\beta E_i)}{\sum \exp(-\beta E_i)}$$

$$= - d \ln(Z) / d\beta$$

Partition Sum & Free Energy:

$$\left. \begin{aligned} Z &\equiv \sum_i \exp(-\beta \epsilon_i) \\ F &= -k_B T \ln(Z) \end{aligned} \right\}$$


$p_i = \exp(-\beta \epsilon_i) / Z$

$$\begin{aligned} S &= -k_B \sum_i p_i \ln p_i = k_B \sum_i p_i (\ln Z + \beta \epsilon_i) \\ &= k_B \ln Z + \langle E \rangle / T = \frac{\langle E \rangle - F}{T} \end{aligned}$$

$$F = k_B T \ln(Z) = \langle E \rangle - T S$$

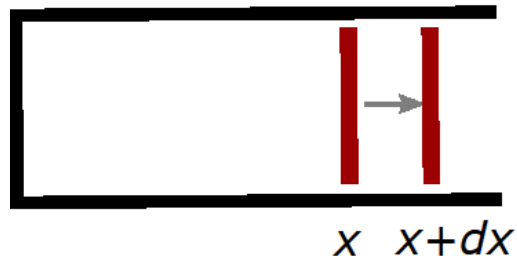
Interpretation, Free energy:

$$F \equiv -k_B T \ln Z = \langle E \rangle - T \langle S \rangle$$

Free energy is energy that is available to do work on surroundings,
Because Z effectively counts probability to be in certain states,
 Z (or F) governs flow toward the most likely state

Z is statistical weight (effective size of available phase space given system is at fixed temperature)

Statistical weight of all states at position x is $Z(x)$



Statistical weight of all states at position $x+dx$ is $Z(x+dx)$

When allowing random moves probability to go



is proportional to relative size of $Z(x+dx)$ and $Z(x)$



$$\text{Force} = d \ln(Z) / dx$$

Partition function (used for constant T)
is actually about partitioning:

$$p(E_1) = \frac{\Omega_1(E_1)\Omega_2(E - E_1)}{\sum \Omega_1(E_1)\Omega_2(E - E_1)}$$



$$p(E_1) = \frac{1}{Z} \Omega_1(E_1) \exp(-E_1/k_B T)$$

$\Omega_2(E - E_1) = \exp\left(\frac{S_2(E - E_1)}{k_B}\right)$

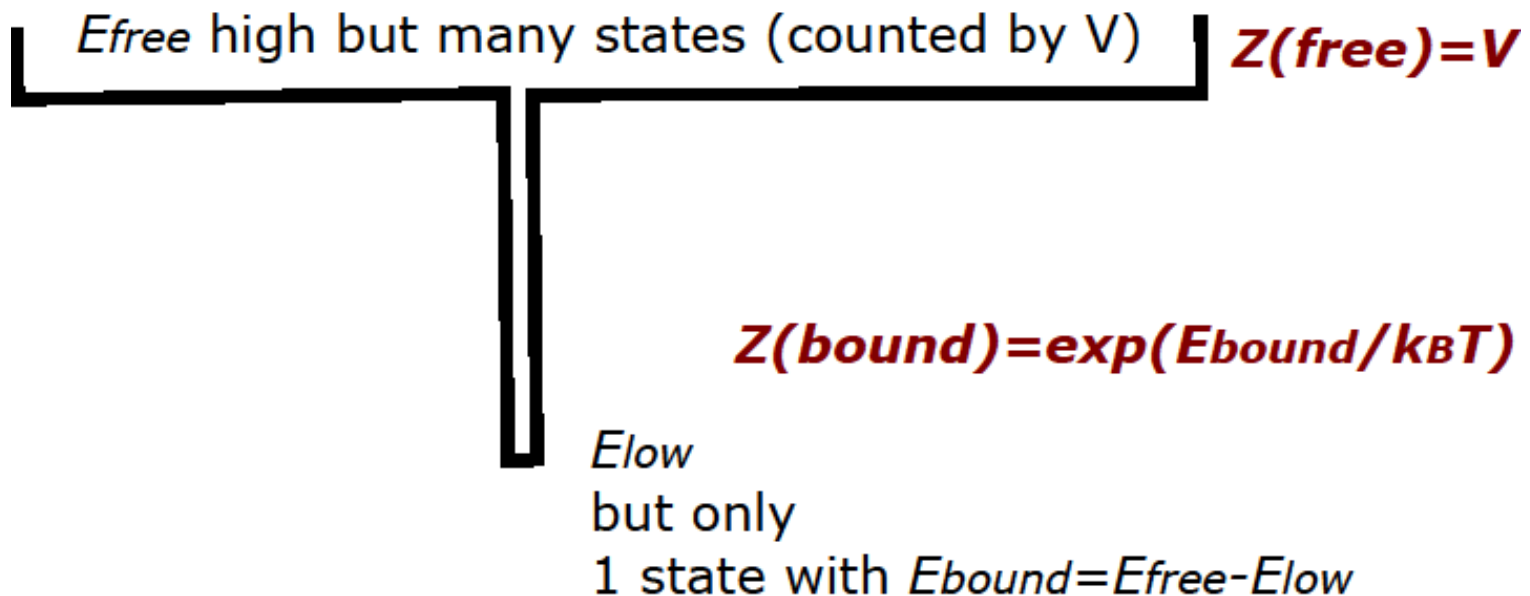
$$\begin{aligned}
 Z &= \sum \Omega_S(E_1) \exp(-E_1/k_B T) \\
 &= \sum_i e^{S(E_i)/k_B} \cdot e^{-E_i/k_B T} \\
 &= \sum_i e^{-F_i/k_B T} = \exp(-F/k_B T)
 \end{aligned}$$

Where index i refers to different macrostates,
each having a free energy

Interpretation Partition function=

Number of states, Boltzmann weighted

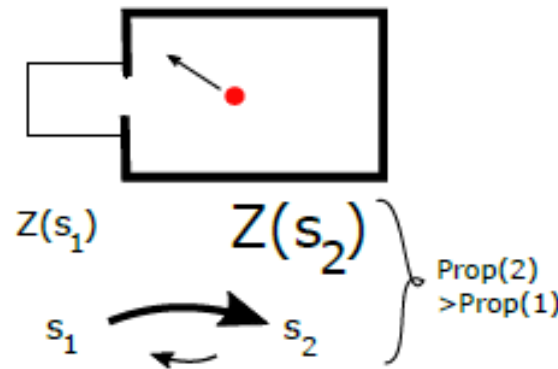
Z is statistical weight, can be partitioned, Example:



The probability of one relative to the other given by $Z(\text{bound})$ versus $Z(\text{free})$

Free energy statements:

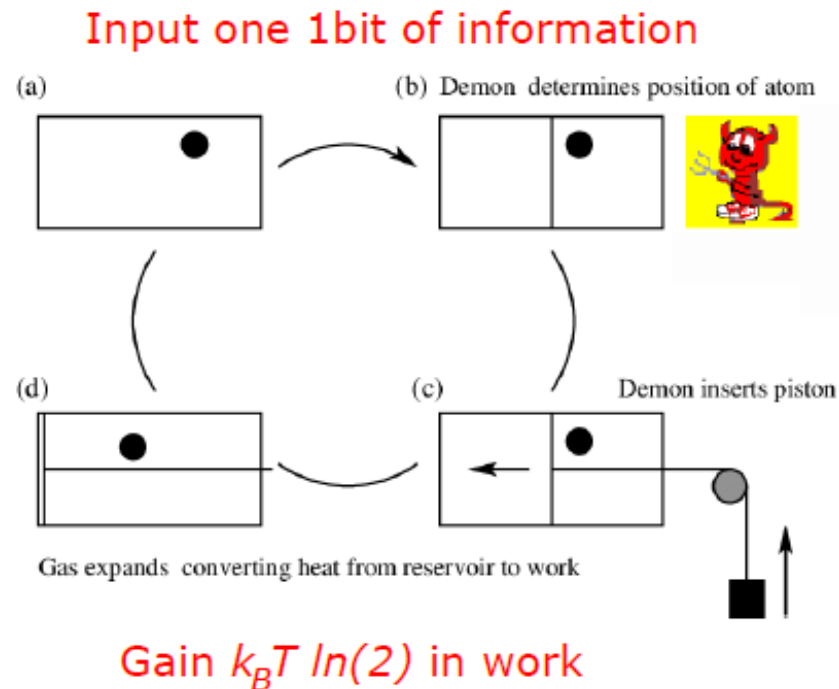
$$F \equiv -k_B T \ln Z = \langle E \rangle - T \langle S \rangle$$



- Free energy is the available energy to do work
- The free energy never increases (for the isolated system at constant T)
- Free energy is minimal at equilibrium.

Use information to extract work from a heat reservoir.

Szilard engine:



Example: Paramagnet, h =external field

microcanonical calculation

$$\mathcal{H} \equiv -h \sum_{i=1}^N s_i \quad s_i = \pm 1;$$

$E = h \cdot n$ with fixed:

$$\begin{aligned} n &= N_{\downarrow} - N_{\uparrow} \\ &= N - 2N_{\uparrow} \end{aligned} \quad \Rightarrow \quad \begin{aligned} N_{\uparrow} &= \frac{N - n}{2} \\ N_{\downarrow} &= \frac{N + n}{2}. \end{aligned}$$

$$\Omega(N, n) = \frac{N!}{(N - N_{\uparrow})! N_{\uparrow}!} = \frac{N!}{\left(\frac{N+n}{2}\right)! \left(\frac{N-n}{2}\right)!}$$

Stirling eq.: $x! = (x/e)^x$

$$\begin{aligned} \frac{S(N, n)}{k_B} &= N \ln N - N - \frac{N+n}{2} \ln \frac{N+n}{2} + \frac{N+n}{2} - \frac{N-n}{2} \ln \frac{N-n}{2} + \frac{N-n}{2} \\ &= N \ln N - \frac{N+n}{2} \ln \frac{N+n}{2} - \frac{N-n}{2} \ln \frac{N-n}{2}. \end{aligned} \quad (1.19)$$

*NB: Can have temperature
In isolated system*

$$\begin{aligned} \frac{1}{T} &= \frac{\partial S(N, E)}{\partial E} = \frac{\partial S(N, n)}{\partial n} \frac{\partial n}{\partial E} \quad (dE = h \cdot dn) \\ &= \frac{1}{2h} \ln \frac{N+n}{N-n}. \end{aligned} \quad \Rightarrow \quad \frac{n(T)}{N} = \frac{E(T)}{Nh} = \tanh \frac{h}{k_B T}.$$

Example: Paramagnet, h =external field

canonical calculation

$S_i=1,2\dots$ independent

$$\begin{aligned} Z_C(T) &= \sum_{s_1=\pm 1, s_2=\pm 1, \dots, s_N=\pm 1} \exp \left(-\frac{h}{k_B T} \sum_{i=1}^N s_i \right) \\ &= \prod_{i=1}^N \left[\sum_{s=\pm 1} e^{-\frac{h}{k_B T} s} \right] = \left[e^{\frac{h}{k_B T}} + e^{-\frac{h}{k_B T}} \right]^N \\ &= 2^N \cdot \cosh \left(\frac{h}{k_B T} \right)^N. \end{aligned}$$

And average energy:

since all spins are equivalent, the total energy is just $N E_j(T)$

$$\begin{aligned} E(T) = N \cdot E_j(T) &= \frac{1}{Z_C(T)} \sum_{s_1=\pm 1, s_2=\pm 1, \dots, s_N=\pm 1} h \cdot s_j \exp \left(-\frac{1}{k_B T} \sum_{i=1}^N h \cdot s_i \right) \\ &= Nh \cdot \frac{\sinh \left(\frac{h}{k_B T} \right) \cdot \cosh \left(\frac{h}{k_B T} \right)^{N-1}}{\cosh \left(\frac{h}{k_B T} \right)^N} \\ &= Nh \cdot \tanh \left(\frac{h}{k_B T} \right), \end{aligned}$$

Partition function $\rightarrow$ Average energy

$$Z \equiv \sum_i \exp(-\beta \epsilon_i)$$

$$\langle E \rangle = \frac{\sum_i \epsilon_i \exp(-\beta \epsilon_i - \alpha N_i)}{Z} = -\frac{\partial}{\partial \beta} \ln Z = T^2 \frac{\partial}{\partial T} \ln Z$$

Paramagnet: $E = -d/d\beta \left(2^N \cdot \cosh \left(\frac{h}{k_B T} \right)^N \right)$

$$\begin{aligned} S &= -k_B \sum_i p_i \ln p_i = k_B \sum_i p_i (\ln Z + \beta \epsilon_i) \\ &= k_B \ln Z + \langle E \rangle / T = \frac{\langle E \rangle - F}{T} \end{aligned}$$

$$C_H \equiv \left(\frac{\partial \langle E \rangle}{\partial T} \right)_H = \frac{\partial \langle E \rangle}{\partial \beta} \frac{d\beta}{dT} = -\frac{1}{k_B T^2} \frac{\partial \langle E \rangle}{\partial \beta} = \frac{1}{k_B T^2} \frac{\partial^2 \ln Z}{\partial \beta^2}$$

And in the case of external magnetization h ($\mathcal{H} \equiv -h \sum_{i=1}^N s_i$) in this could be varied to give:

$$M = - \left(\frac{\partial F}{\partial h} \right)_T$$

$$\chi T = - \left(\frac{\partial^2 F}{\partial h^2} \right)_T$$

Susceptibility=
How much more
magnetism for some
external magnetism

Ising model:

$$\mathcal{H} \equiv -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i \quad s_i = \begin{cases} +1 & \text{"spin up"} \\ -1 & \text{"spin down"} \end{cases}$$

Coupling strength
between spins

External field that try to make
Spins s_i pointing in the direction of h

Only interactions between
neighbors on some lattice

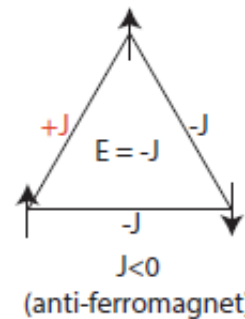
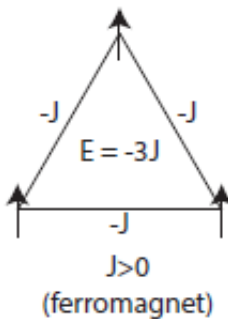
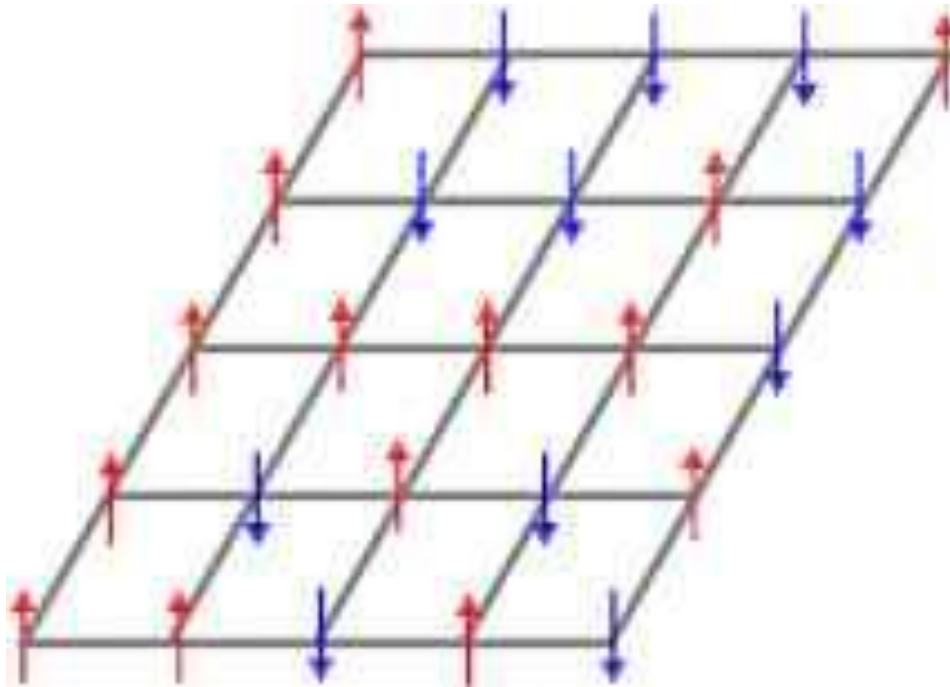


Figure 1.7: Significance of the sign of coupling.

Try to align spins

Try to anti-align spins

Ising model:



Monte Carlo Method

- Aim is to calculate:

$$\langle A \rangle = \frac{\sum_{\{s\}} A \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

- Where the sum Σ is over all states.
- If 100x100 lattice, this is $2^{10000} = 10^{3000}$ states... impossible to access by random sampling.
- Need to be smart.... → **Importance sampling**
- Can do random sampling, or ?

Examples, Want to calculate:

$$\langle s_i \rangle = \langle s \rangle = \frac{\sum_{\{s\}} s_i \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

$$\langle s_i s_j \rangle = \frac{\sum_{\{s\}} s_i s_j \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

$$\langle \mathcal{H} \rangle = \frac{\sum_{\{s\}} \mathcal{H} \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

Want to do this using **Importance sampling**,... i.e.
start sample and continue using gained knowledge to select
new samples that are relevant

Importance sampling

- Markov chain of sampled states to sample $P_l^{eq} = \frac{e^{-\beta E_l}}{\sum_m e^{-\beta E_m}}$

- Probability to be in state l at time t :

$$P_l(t+1) = P_l(t) \left(1 - \sum_{m \neq l} w_{l \rightarrow m} \right) + \sum_{m \neq l} P_m(t) w_{m \rightarrow l} \quad (\text{Where } \sum_m w_{l \rightarrow m} = 1)$$

- Notice that $\sum_l P_l(t+1) = \sum_{m,l} P_m(t) w_{m \rightarrow l} = \sum_{m,l} P_l(t) w_{l \rightarrow m} = \sum_l P_l(t) = 1$
- Stationarity** $P_l(t+1) = P_l(t) \rightarrow \sum_m [P_l^{eq} w_{l \rightarrow m} - P_m^{eq} w_{m \rightarrow l}] = 0$

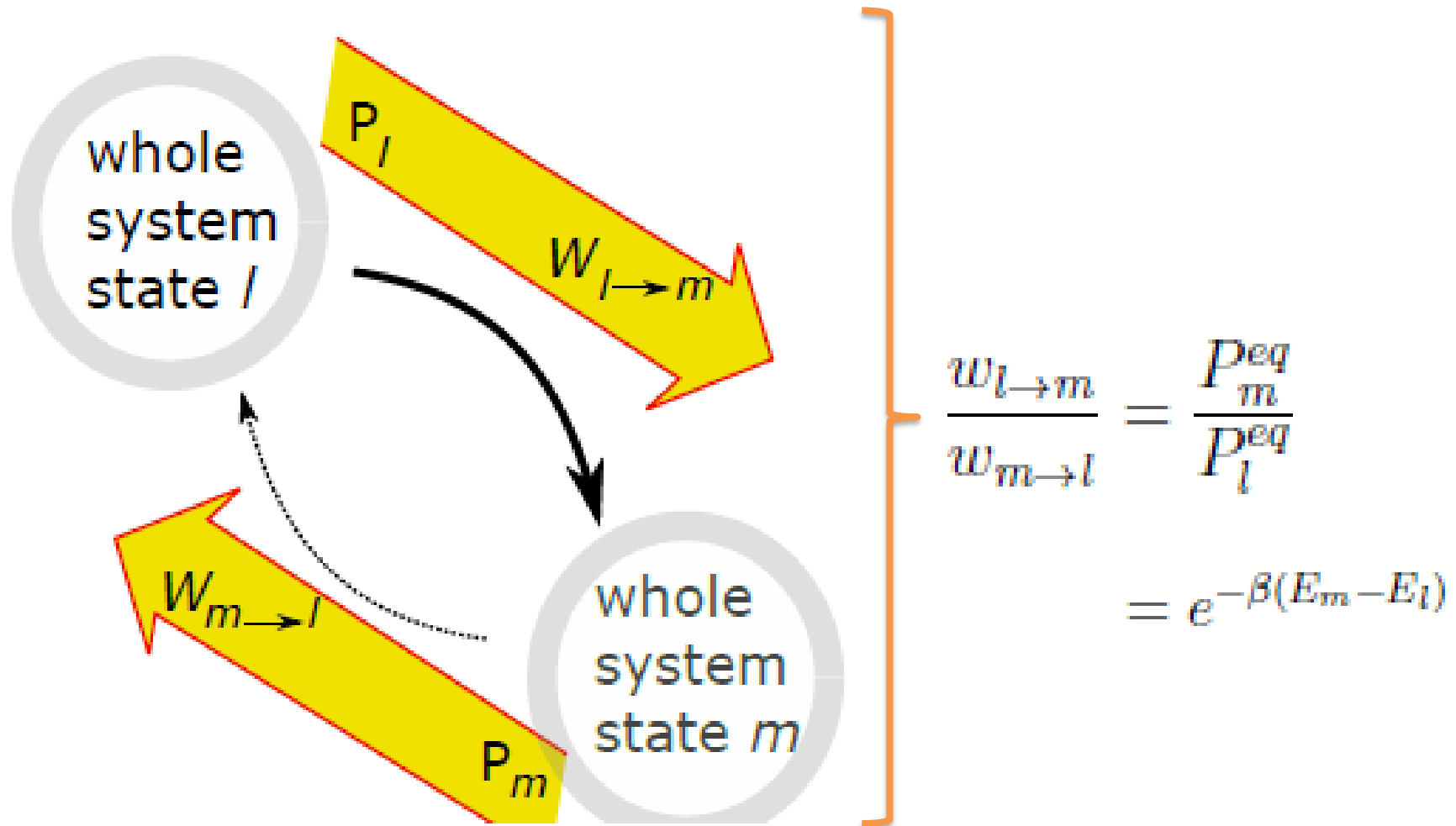
- This is fulfilled if:

$$P_l^{eq} w_{l \rightarrow m} = P_m^{eq} w_{m \rightarrow l}$$

- Or, **detailed balance**: $\frac{w_{l \rightarrow m}}{w_{m \rightarrow l}} = \frac{P_m^{eq}}{P_l^{eq}} = e^{-\beta(E_m - E_l)} \equiv e^{-\beta \Delta E}$

Can calculate averages by sampling, without the need for Z

Detailed balance:



Standard sampling:

We make the ansatz that the transition probabilities should be a function of only energy differences, i.e. that $w_{l \rightarrow m}$ should be a function of $e^{-\beta \Delta E}$:

$$w_{l \rightarrow m} = F(e^{-\beta \Delta E}) ,$$

Thereby we don't need to calculate the total energy with a move, but only need to calculate the difference for the move.

by symmetry hence

$$w_{m \rightarrow l} = F\left(\frac{1}{e^{-\beta \Delta E}}\right) \equiv F\left(\frac{1}{x}\right) \quad \text{with } x \equiv e^{-\beta \Delta E}$$

Detail balance requirement then implies that

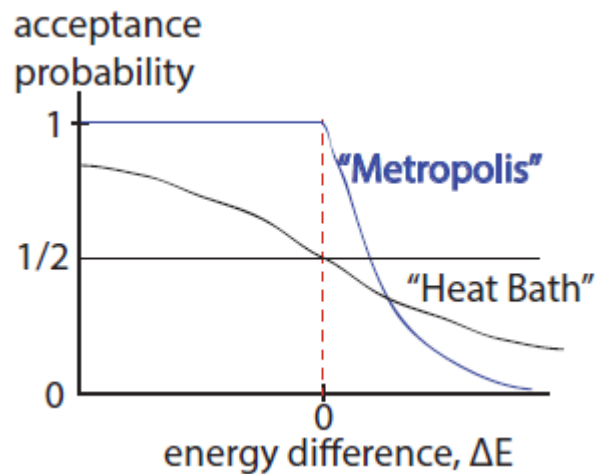
$$\frac{w_{l \rightarrow m}}{w_{m \rightarrow l}} = \frac{F(x)}{F(1/x)} = x$$

Furthermore it must be ensured that $0 \leq F(x) \leq 1$

Anything that fulfills this is ok

Simplest choice:

- $F(x)=x$ if $x<1$
- $F(x)=1$ if $x\geq 1$



$F(x) = \min(x, 1)$, Metropolis algorithm [3, 4]. This means that one always accept a suggested change in state that lower energy, but only accept a increase in energy with a probability given by the Boltzman factor (see Figure).

Practical implementation:

If $\Delta E < 0$: always accept

If $\Delta E > 0$: accept if random $< \exp(-\Delta E/k_B T)$

Where random is uniformly selected in $[0, 1]$.

$F(x) = \frac{x}{1+x}$, heat bath algorithm.

Proof: $F(x)/F(1/x) = [x/(x+1)] [(1+1/x)/(1/x)] = x$

The $\frac{1}{2}$ is to fulfill $F(x) \leq 1$.

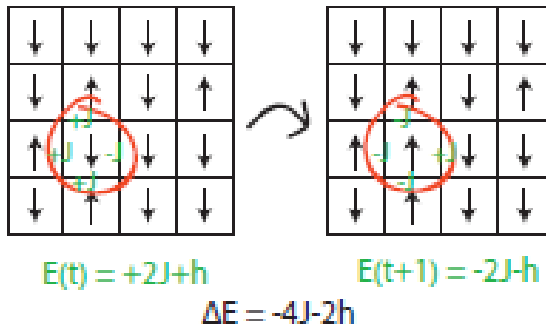


$F(x) = \frac{1}{2} (1 - \tanh(\beta \Delta E / 2))$, Glauber dynamics.

Proof: $F(x)/F(1/x) = \dots = \text{complete yourself}$

Practical use of Metropolis:

$$\mathcal{H} \equiv -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i$$



1 spin aligned $\rightarrow$ 3 spin aligned.

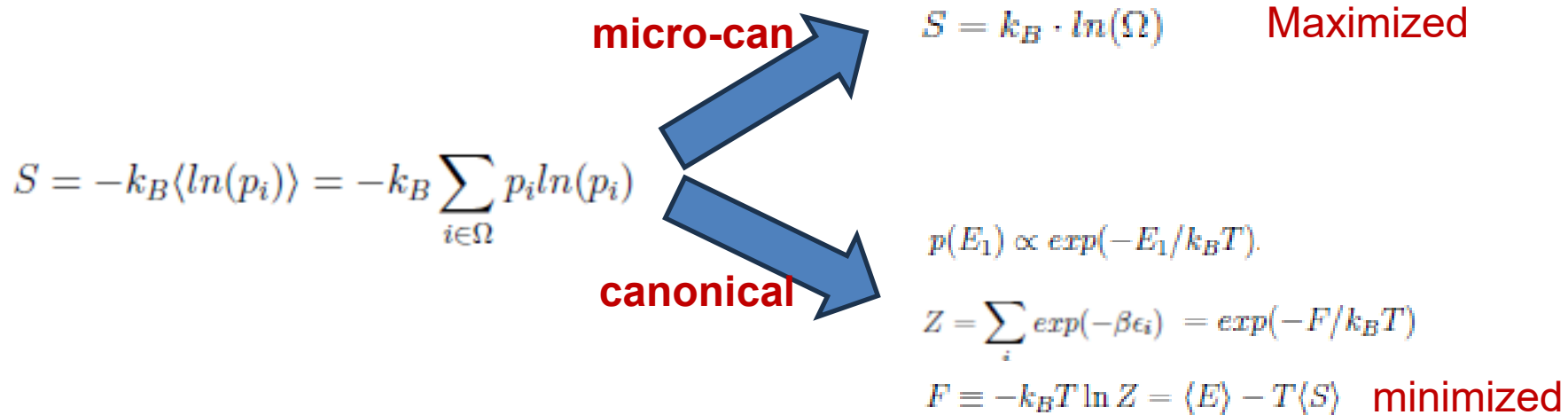
If $h > 2J$ then the move is accepted.

If $h < -2J$ then move only happens if:

$\leftarrow$ Accepted if $random < \exp(-(-4J - 2h)/k_B T)$
 i.e. $< \exp[(4J + 2h)/k_B T]$

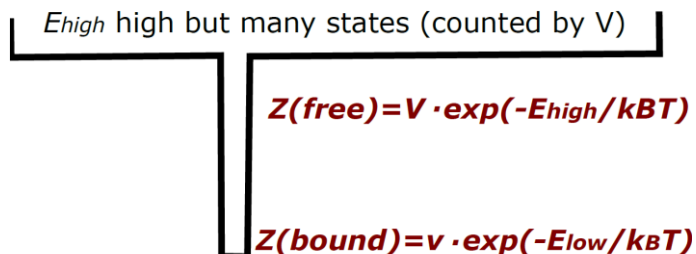
Thursday the 4'th September

Recapitulation /equilibrium:



$p(E_1) \propto \exp(-E_1/k_B T)$ **Caused by constrain on entropy of neighbors**

Z is statistical weight, can be partitioned in macrostates, for example:



E_{low} but only few states v
 with $E_{bound} = E_{high} - E_{low}$

Two parts partition sum, often many parts, fex:

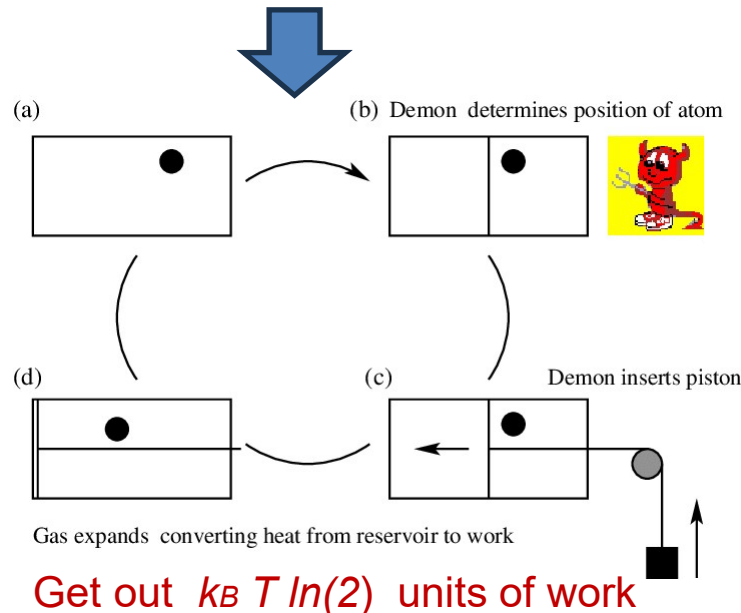
$$Z = \sum_i e^{S(E_i)} \cdot e^{-E_i/k_B T} = \sum_i e^{-F_i/k_B T} = \exp(-F/k_B T)$$

Recapitulation (4/9 2024):

- Differences in Free energy is the available energy to do work
- The free energy never increases (for the isolated system at constant T)
- Free energy is minimal at equilibrium.

$F = E - TS$ contains both E and S, the combination can be used to extract work.
Szilard engine uses a daemon to reduce S and extract work:

Put in 1 bit of information ($\rightarrow$ lower S, increases F)



Question:
Where does
energy comes
From?

Recap: Monte Carlo Method

- Aim is to calculate:

$$\langle A \rangle = \frac{\sum_{\{s\}} A \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

- Where the sum Σ is over all states.

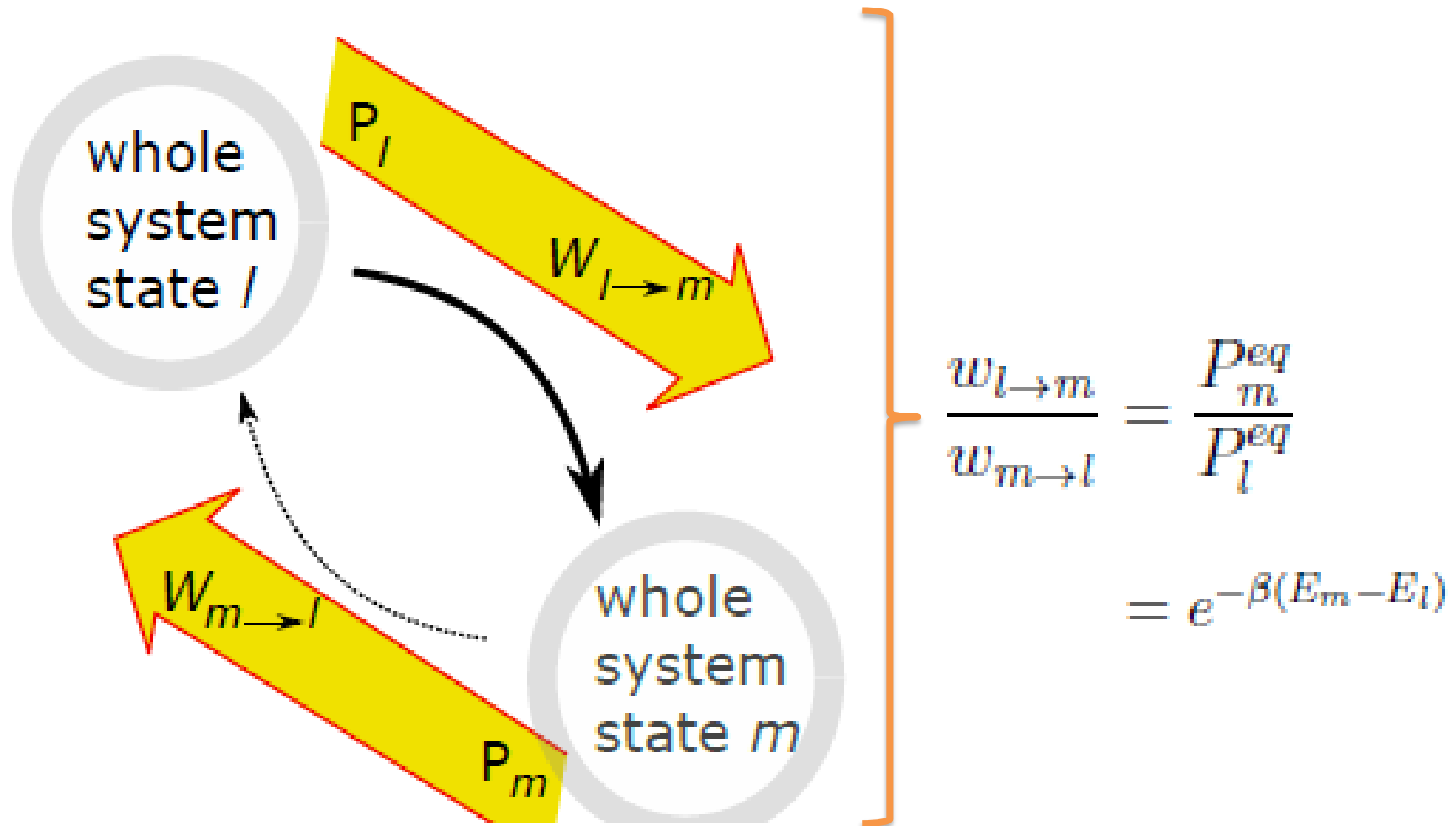
Recap; F.ex. want to calculate:

$$\langle s_i \rangle = \langle s \rangle = \frac{\sum_{\{s\}} s_i \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

$$\langle s_i s_j \rangle = \frac{\sum_{\{s\}} s_i s_j \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

$$\langle \mathcal{H} \rangle = \frac{\sum_{\{s\}} \mathcal{H} \exp(-\beta \mathcal{H})}{\sum_{\{s\}} \exp(-\beta \mathcal{H})}$$

Detailed balance:



Simplest approach:

$$\frac{w_{l \rightarrow m}}{w_{m \rightarrow l}} = \frac{P_m^{eq}}{P_l^{eq}} = e^{-\beta(E_m - E_l)}$$

Practical implementation:

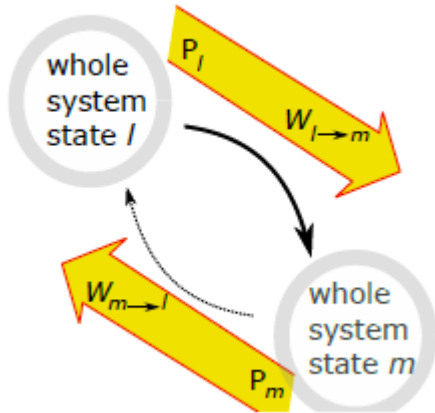
If $\Delta E < 0$: always accept

If $\Delta E > 0$: accept if random $< \exp(-\Delta E/k_B T)$

Where random is uniformly selected in $[0, 1]$.

Recapitulation (10/9-2024):

Importance sampling:



$$\frac{w_{l \rightarrow m}}{w_{m \rightarrow l}} = \frac{P_m^{eq}}{P_l^{eq}} = e^{-\beta(E_m - E_l)} \equiv e^{-\beta \Delta E}$$

Uphill if $ran < \exp(-\Delta E/k_B T)$
Downhill always accepted

N_0 =transient steps
 N_{max} =simulation steps

$$\langle A \rangle = \frac{1}{n_{max} - n_0} \sum_{n > n_0} A_n$$

Faster to average over some of these states

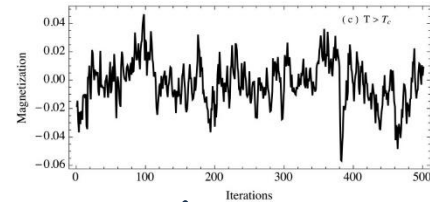
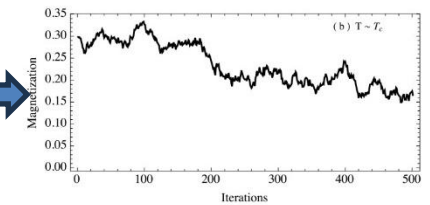
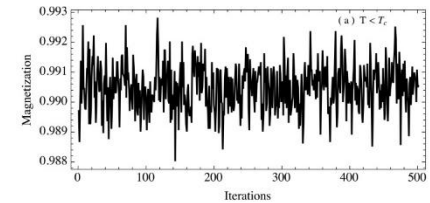
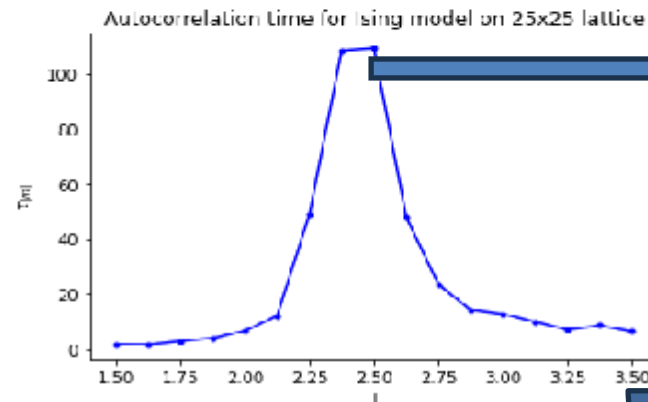
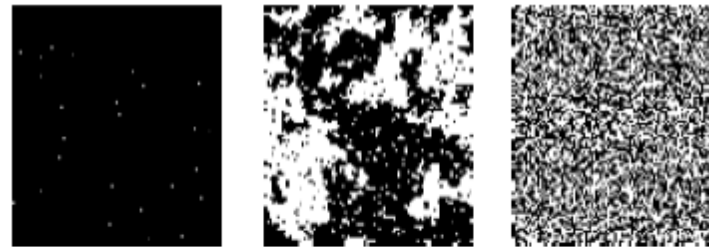
Practical implementation of Metropolis (for the Ising model):

The basic setup of a Monte Carlo simulation is rather straightforward:

- Set up lattice sites i and spins s_i , define Hamiltonian $\mathcal{H}$, define total number of steps n_{max} and $n_0 < n_{max}$,
- Set the system to an initial configuration of spins (either a random configuration or one that is meaningful for the problem of interest),
- Flip a spin, compute $r \equiv e^{-\Delta E/k_B T}$ (compare Eq. 1.36), generate a uniformly-distributed real random number q within the interval $[0, 1]$, if $r > q$ perform the Monte Carlo move (e.g. flip the spin), otherwise do not,
- Compute A_n , i.e. the expectation value of A at time step n , if $n > n_0$, if $n \leq n_0$, we are still in the period considered transient,
- Repeat until n_{max} is reached,
- Calculate $\langle A \rangle = \frac{1}{n_{max} - n_0} \sum_{n > n_0} A_n$. In practice, you could speed up the simulation by only averaging between some of the sampled states.

Difficulties usually lie in the proper choice of system size N , the choice of the transient period n_0 , and the duration of the sampling period n_{max} . Generally, fluctuations increase in the vicinity of T_c and achieving robust results may require increasing both system size and sampling time. Approaches on how to choose the values of n_{max} and n_0 more systematically can be found in the literature [6, 7].

Warning: Sometimes lo..oo..ong transients

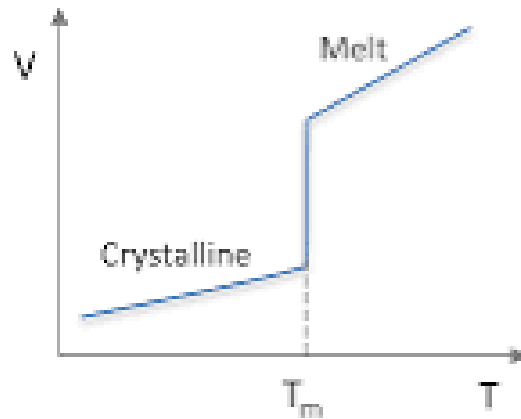


$$\tau \equiv \frac{\int_0^\infty t \cdot [A(t) - \langle A \rangle] dt}{\int_0^\infty [A(t) - \langle A \rangle] dt} \sim \xi^z \sim |T_c - T|^{-z\nu}$$

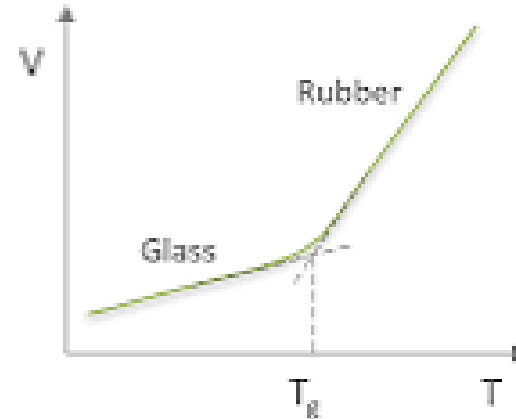
Only get independent value after τ updates!

Phase Transitions

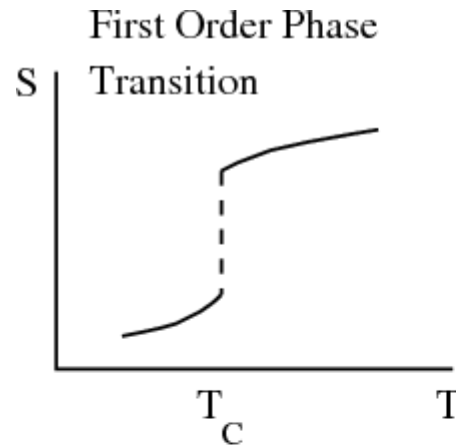
First Order Transition



Second Order Transition

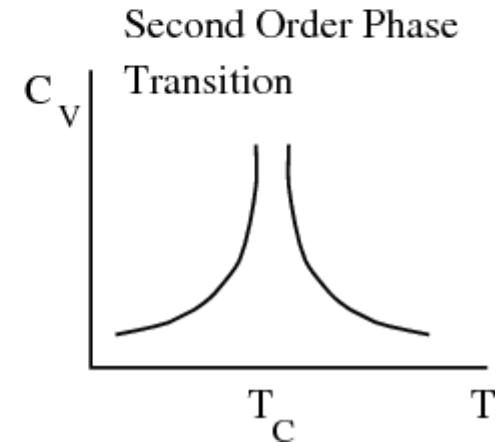
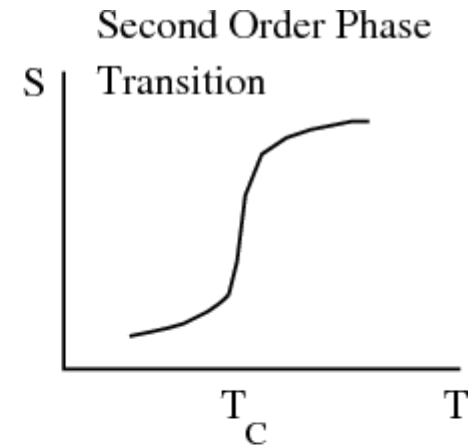


Phase Transitions:



$$L = T\Delta S > 0$$

($F=E-TS$ continuous
 $\rightarrow$ Latent heat
 $L=\Delta E=T\Delta S$)



$$L = T\Delta S = 0$$

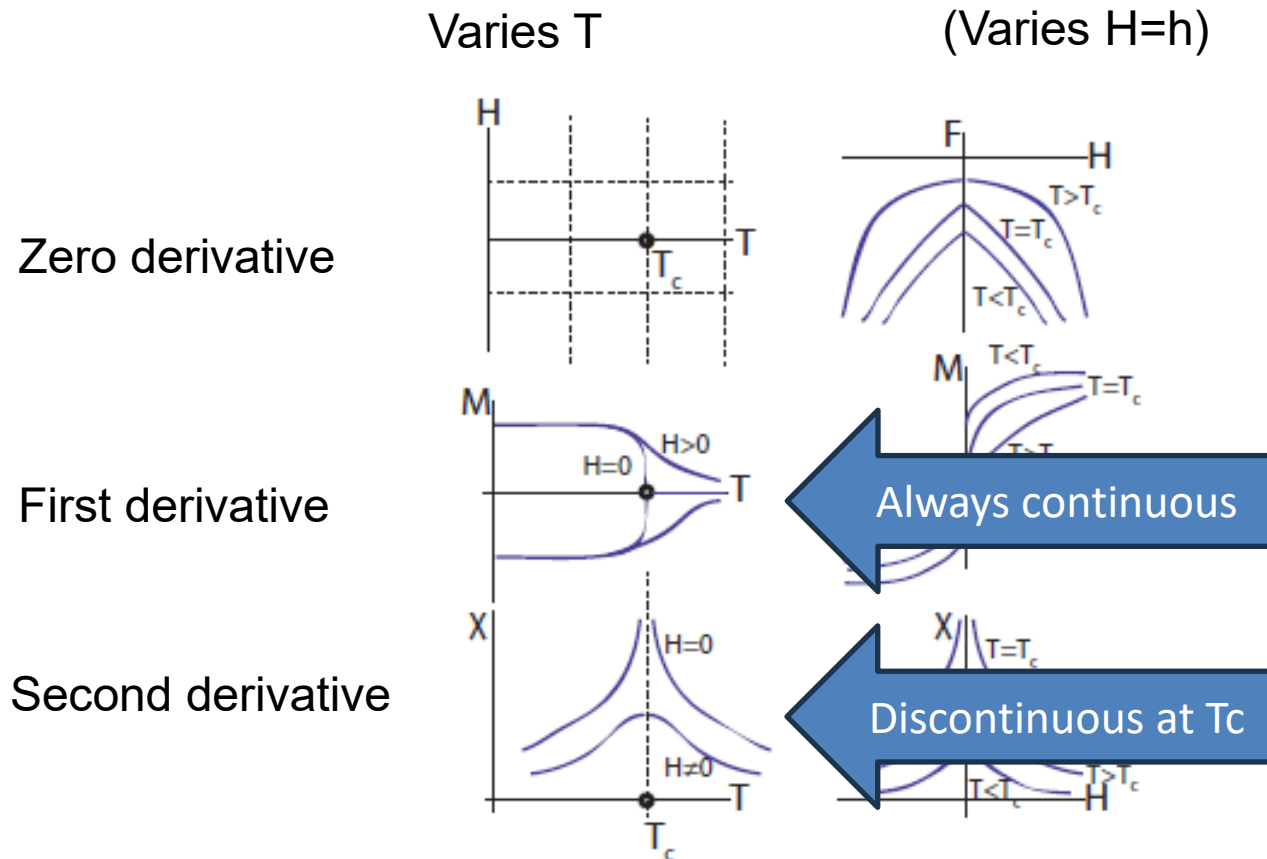
Phase Transitions & Criticality

$$\mathcal{H} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i$$

- **Order parameter:** Magnetization $m = \frac{1}{N} \sum_i s_i$
- Liquid-gas transition: density $\Delta\rho = \rho_{\text{liquid}} - \rho_{\text{gas}}$

Average magnetization: $m = \frac{1}{N} \frac{d \ln(Z)}{dh} = \frac{\sum_i h s_i \exp(-\beta \mathcal{H})}{\sum_i \exp(-\beta \mathcal{H})}$

Second order transition in Ising model



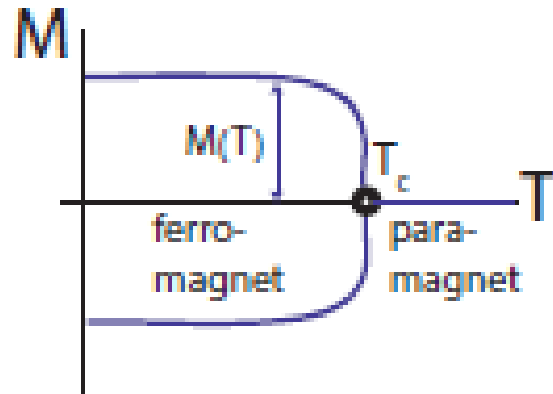
Susceptibility=
How much more
magnetism for some
external magnetism

$$M = - \left(\frac{\partial F}{\partial h} \right)_T$$

Figure 2.3: Dependencies near the critical point.

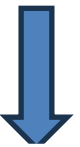
$$\chi_T = - \left(\frac{\partial^2 F}{\partial h^2} \right)_T$$

Correlation length:



$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$ $T=0$ (ground state)
 $\uparrow\uparrow\uparrow\uparrow\downarrow\uparrow\uparrow\downarrow\uparrow\uparrow\uparrow\uparrow$ $T < T_c$ (e.g. $M > 0$)
 $\downarrow\downarrow\uparrow\downarrow\downarrow\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow$ $T = T_c$ ($M = 0$)
 $\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow$ $T > T_c$ ($M = 0$)

Correlation length



All points equal $\rightarrow$ correlation length diff. $\rightarrow$

$$\Gamma(\mathbf{r}_i, \mathbf{r}_j) = \langle (s_i - \langle s_i \rangle)(s_j - \langle s_j \rangle) \rangle \rightarrow \Gamma(\mathbf{r}_i - \mathbf{r}_j) \equiv \Gamma_{ij} = \langle s_i s_j \rangle - \langle s \rangle^2 \rightarrow \Gamma(\mathbf{r}) \sim r^{-\tau} \exp(-r/\xi)$$

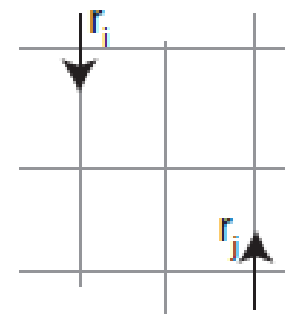
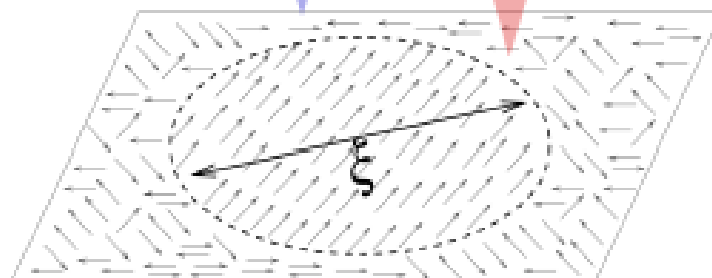
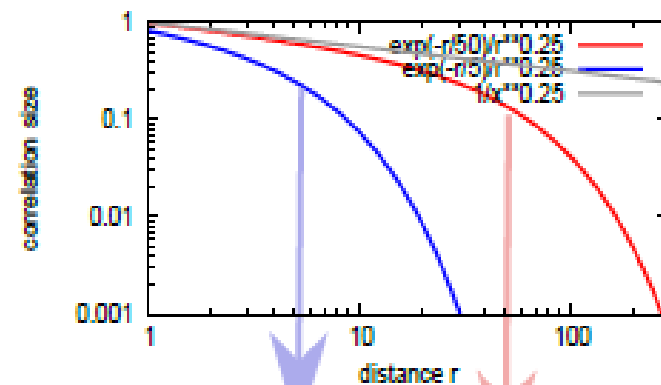
$$\xi \propto 1/(T_c - T)$$

$$\Gamma(\mathbf{r}) \sim r^{-\tau} \exp(-r/\xi)$$

$$\Gamma(\mathbf{r}) \sim \frac{1}{r^{d-2+\eta}}$$

$$= 1/r^{0.25}$$

$$\xi \propto 1/(T_c - T)$$



Magnetic susceptibility & spin-spin space correlations

$$M = (1/\beta) \cdot \partial \ln(Z) / \partial h$$

$$M = - \left(\frac{\partial F}{\partial h} \right)_T$$

$$\begin{aligned} \chi_T &= k_B T \frac{\partial^2}{\partial H^2} \ln Z \\ &= \frac{1}{k_B T} (\langle M^2 \rangle - \langle M \rangle^2) \\ &= \frac{1}{k_B T} \langle (M - \langle M \rangle)^2 \rangle \\ &= \frac{1}{k_B T} \langle \sum_i (s_i - \langle s_i \rangle) \sum_j (s_j - \langle s_j \rangle) \rangle \\ &= \frac{1}{k_B T} \sum_{ij} \Gamma_{ij} = N \sum_j \Gamma_{0j} \sim N \int dr \Gamma(r) r^{d-1} \end{aligned}$$

$$\chi_T = - \left(\frac{\partial^2 F}{\partial h^2} \right)_T = dM/dh$$

$$\xi^2 / \xi^{0.25} = 1 / (T_c - T)^{1.75}$$

with d=dimension of lattice, N=number of lattice points and

$$\Gamma(\mathbf{r}) \sim r^{-\tau} \exp(-r/\xi)$$

$$\Gamma(\mathbf{r}) \sim \frac{1}{r^{d-2+\eta}}$$

$$= 1/r^{0.25}$$

$$\xi \propto 1/(T_c - T)$$

Contributes out to infinity for $T=T_c \rightarrow$ susceptibility diverges due to *spin-spin* correlations.

Critical exponents

$$t \equiv \frac{T - T_c}{T_c}$$

$C_H \sim |t|^{-\alpha}$ zero-field specific heat

$M \sim (-t)^\beta$ zero-field magnetization

$\chi_T \sim |t|^{-\gamma}$ zero-field isothermal susceptibility

$H \sim |M|^\delta \text{sgn}(M)$ critical isotherm ($t = 0$)

$\xi \sim |t|^{-\nu}$ correlation length

$G(\mathbf{r}) \sim \frac{1}{r^{d-2+\eta}}$ pair-correlation function at T_c .

$\gamma = d - \tau$, $d=2$ or 3

$$\Gamma(\mathbf{r}) \sim r^{-\tau} \exp(-r/\xi)$$

C M χ $H(M)$ ξ $\Gamma(r)$

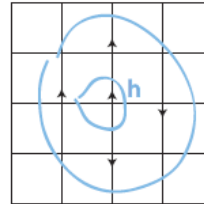
Universality class	symmetry	α	β	γ	δ	ν	η
2D Ising	2-component scalar	0	1/8	7/4	15	1	1/4
3D Ising	2-component scalar	0.10	0.33	1.24	4.8	0.63	0.04
3D XY	2-dimensional vector	0.01	0.34	1.3	4.8	0.66	0.04
3D Heisenberg	3-dimensional vector	-0.12	0.36	1.39	4.8	0.71	0.04
mean field		0	1/2	1	3	1/2	0
2D Potts, $q = 3$	q -component scalar	1/3	1/9	13/9	14	5/6	4/15

Table 2.1: Critical exponents for several models.

They are **universal**, independent of (model) details
(like the type of lattice doesn't matter)

Mean Field Solution

$$\mathcal{H} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i$$



For each site replace all interacting neighbor spins with the average spin across the lattice

$$m = \frac{1}{N} \sum_{j=1}^N \langle s_j \rangle \quad s_i = m + (s_i - m)$$

$$\begin{aligned} s_i s_j &= (m + (s_i - m))(m + (s_j - m)) \\ &= m^2 + m(s_i - m) + m(s_j - m) + (s_i - m)(s_j - m) \end{aligned}$$

ignore



$$\begin{aligned}
 s_i s_j &= (m + (s_i - m))(m + (s_j - m)) \\
 &= m^2 + m(s_i - m) + m(s_j - m) + (s_i - m)(s_j - m)
 \end{aligned}$$

ignore

$$E_{MF} = -J \sum_{\langle ij \rangle} (-m^2 + m(s_i + s_j)) - h \sum_i s_i$$

Factor 1/2
from double
counting

$$-J \sum_{\langle ij \rangle} (-m^2) = \frac{JNz}{2} m^2$$

$$\sum_{\langle ij \rangle} (s_i + s_j) = z \sum_j s_j$$

Where z=4 for square lattice

$$E_{MF} = \frac{JNz}{2} m^2 - (Jzm + h) \sum_{j=1}^N s_j$$

Mini Tutorial: Consider the mean field energy above and discuss its dependence on dimensionality and geometry of the lattice.

$$m = -\frac{1}{N} \frac{d \ln(Z)}{dh} = \frac{1}{N} \sum_{j=1}^N \langle s_j \rangle = \langle s_i \rangle \equiv \langle s \rangle$$

$$\begin{aligned}
Z_{MF} &= \sum_{\{s_j\}} \exp(-\beta E_{MF}) \\
&= \exp\left(-\beta \frac{NJzm^2}{2}\right) \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \prod_{j=1}^N \exp(\beta(Jzm + h)s_j) \\
&= \exp\left(-\beta \frac{NJzm^2}{2}\right) \left[\sum_{s_1=\pm 1} e^{\beta(Jzm+h)s_1} \right] \cdot \dots \cdot \left[\sum_{s_N=\pm 1} e^{\beta(Jzm+h)s_1} \right] \\
&= \exp\left(-\beta \frac{NJzm^2}{2}\right) [e^{\beta(Jzm+h)} + e^{-\beta(Jzm+h)}] \cdot \dots \cdot [e^{\beta(Jzm+h)} + e^{-\beta(Jzm+h)}] \\
&= \exp\left(-\beta \frac{NJzm^2}{2}\right) [2 \cosh(Jzm\beta + h\beta)]^N .
\end{aligned}$$

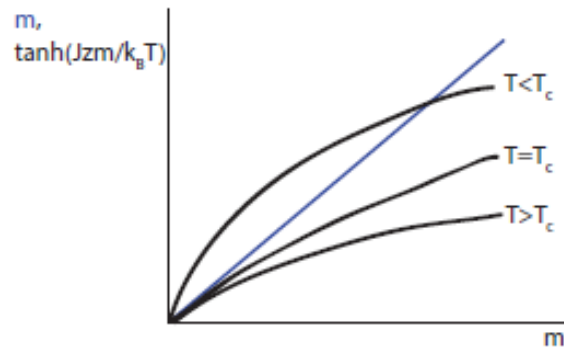
It is now straightforward to compute the magnetization per site:

$$m = -\frac{1}{N} \frac{d \ln(Z)}{dh} = \frac{1}{N} \sum_{j=1}^N \langle s_j \rangle = \langle s_i \rangle \equiv \langle s \rangle ,$$

where naturally value $\langle s_i \rangle$ for any site $i \in \{1, \dots, N\}$ is equal. This equation could also be obtained from exploring minima in Free energy $dF/dm = 0$ (or $d \ln(Z)/dm = 0$). Hence

$$\begin{aligned}
m &= \frac{\exp\left(-\beta \frac{NJzm^2}{2}\right) [2 \cosh(Jzm\beta + h\beta)]^{N-1} [2 \sinh(Jzm\beta + h\beta)]}{\exp\left(-\beta \frac{NJzm^2}{2}\right) [2 \cosh(Jzm\beta + h\beta)]^N} \\
&= \tanh(Jzm\beta + h\beta) .
\end{aligned}$$

Mean Field Solution



$$m = \tanh(Jzm\beta)$$



$$T_c \equiv \frac{Jz}{k_B}$$

Mean field free energy:

$$h' \equiv h/zJ$$

$$\theta = T/T_c = k_B T/Jz$$

$$t \equiv \frac{T - T_c}{T_c} = \theta - 1$$

Mean field critical exponents, $T < T_c$:

$$\tanh(x) = x - \frac{x^3}{3} + \mathcal{O}(x^5)$$

Below T_c we use $m = \tanh[\beta(h + qJm)]$ Which with $h=0$ gives:

$$m = \beta q J m - \frac{1}{3} (\beta q J m)^3$$

$$= \left(\frac{T_c}{T}\right) m - \frac{1}{3} \left(\frac{T_c}{T}\right)^3 m^3, \quad T \rightarrow T_c^-, \quad \Rightarrow \quad m \left[\left(\frac{T_c}{T} - 1\right) - \frac{1}{3} \left(\frac{T_c}{T}\right)^3 m^2 \right] = 0, \quad T \rightarrow T_c^-.$$

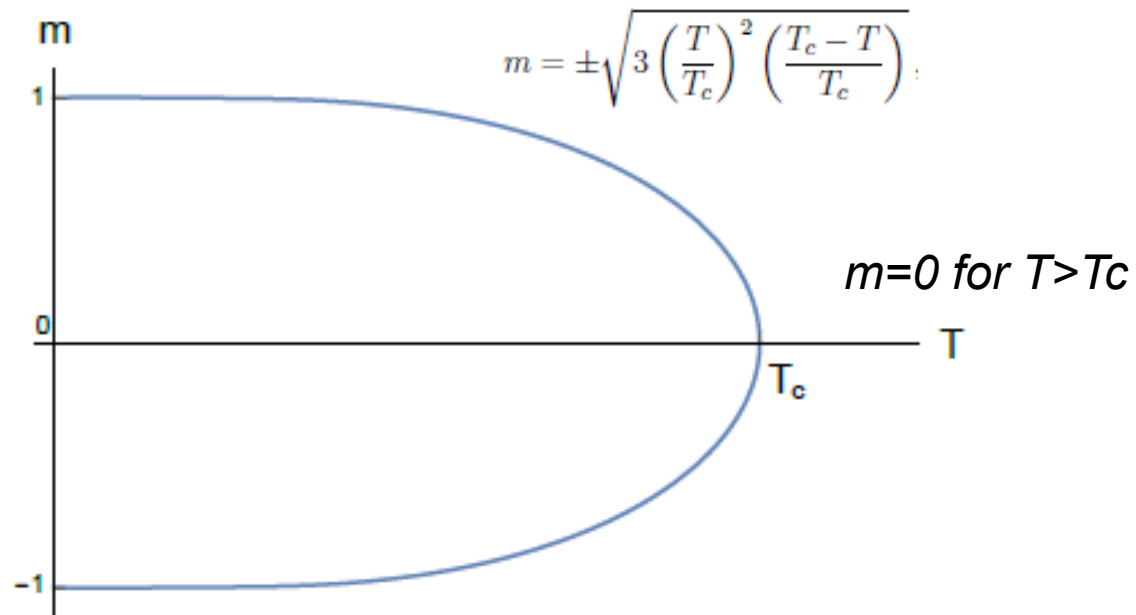
$$\Rightarrow m = \pm \sqrt{3 \left(\frac{T}{T_c}\right)^2 \left(\frac{T_c - T}{T_c}\right)}.$$

$$\Rightarrow m(T, 0) = \begin{cases} \pm (3|t|)^{1/2}, & T \rightarrow T_c^- \\ 0, & T \rightarrow T_c^+ \end{cases}$$

$\beta = 1/2$:

Can also be obtained by expansion & $dF/dm=0$, i.e.: $\frac{1}{zJ} \frac{\partial f_{MF}}{\partial m} = m \left(1 - \frac{1}{\theta}\right) + \frac{1}{3} \frac{m^3}{\theta^3} = 0$

Mean Field behavior



$$m = \tanh[\beta(h + qJm)] \quad \text{Expansion in } h:$$

$$m = \tanh(\beta J z m) + h\beta(1 - \tanh^2(\beta J z m))$$

$$\chi_T(T, h) \equiv \left(\frac{\partial m}{\partial h} \right)_T \quad \longrightarrow \quad m = \tanh(\beta J z m) + h\beta(1 - \tanh^2(\beta J z m))$$

$$\tanh(x) = x - \frac{x^3}{3} + \mathcal{O}(x^5)$$

$$\longrightarrow \quad m = \beta J z m - \frac{1}{3}(\beta J z m)^3 + h\beta - h\beta(\beta J z m)^2$$

NB let $h \rightarrow 0$,
terms with h vanish

$$T_c \equiv \frac{Jz}{k_B}$$

$$\begin{aligned} \chi_T &= \left. \frac{\partial m}{\partial h} \right|_{h=0} \\ &= \beta J z \chi_T - (\beta J z)^3 m^2 \chi_T + \beta - \beta(\beta J z)^2 m^2 \end{aligned}$$

$$m = \pm \sqrt{3 \left(\frac{T}{T_c} \right)^2 \left(\frac{T_c - T}{T_c} \right)}$$

$$\longrightarrow \quad \chi_T \left(1 - \frac{T_c}{T} - \left(\frac{T_c}{T} \right)^3 3t \right) = \frac{1}{kT} \left(1 + 3 \left(\frac{T_c}{T} \right)^2 t \right)$$

$$\lim_{t \rightarrow 0, t < 0} \chi_T = \frac{1}{k_B T_c} \frac{1}{(-2t)}$$

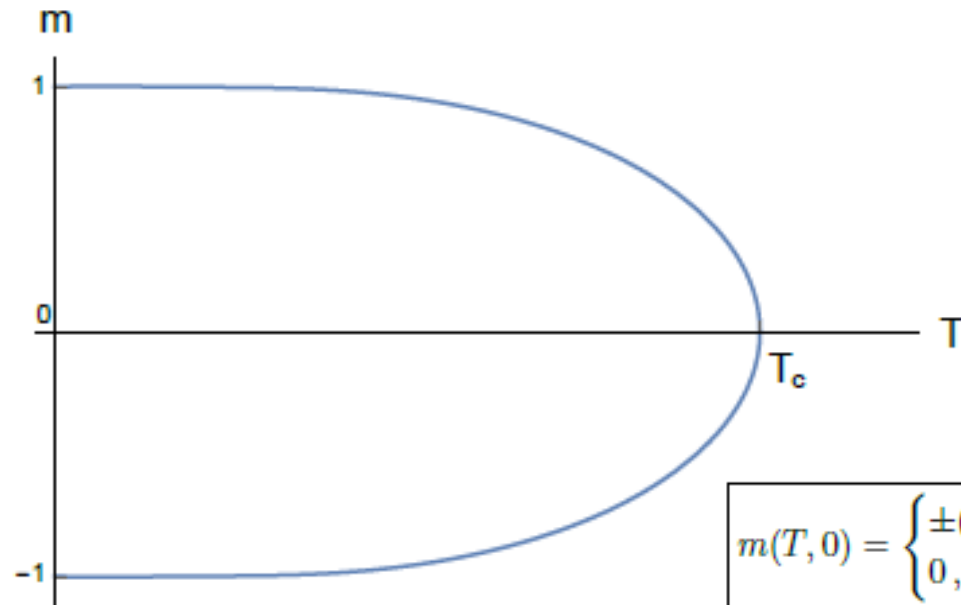
$$\lim_{t \rightarrow 0, t > 0} \chi_T = \frac{1}{k_B T_c} \frac{1}{t} = \frac{1}{Jz t}$$

$$1/|T_c - T|$$

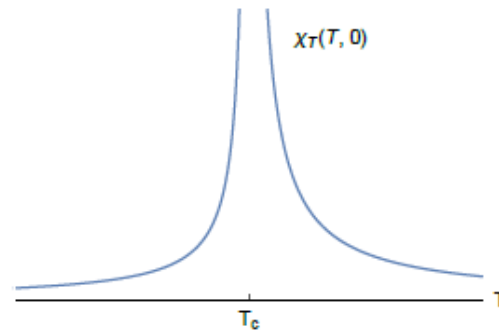
$$\gamma = 1:$$

$$T > T_c \rightarrow m = 0$$

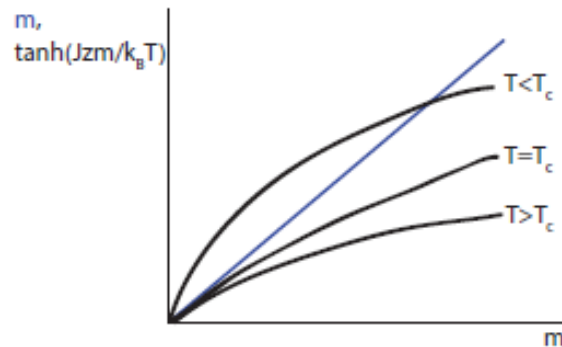
Mean Field behavior



$$\chi_T(T, 0) = \frac{\beta}{1 - \beta q J} = \frac{1}{k_B(T - T_c)} = \frac{1}{k_B T_c} \frac{1}{|t|}, \quad T \rightarrow T_c^+$$



Mean Field Solution



$$m = \tanh(Jzm\beta) \quad \Rightarrow \quad T_c \equiv \frac{Jz}{k_B}$$

Mean field free energy:

$$f_{MF} = -\frac{k_B T}{N} \ln Z_{MF} = \frac{Jz m^2}{2} - k_B T \ln [2 \cosh(Jzm\beta + h\beta)]$$

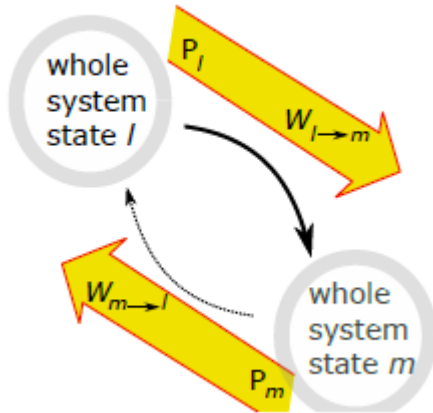
$$\frac{f_{MF}}{zJ} = \frac{m^2}{2} - \theta \ln 2 - \theta \ln \cosh \left(\frac{m + h'}{\theta} \right)$$

$$h' \equiv h/zJ$$

$$\theta = T/T_c = k_B T / Jz$$

$$t \equiv \frac{T - T_c}{T_c} = \theta - 1$$

Re-Recapitulation (8/9), Importance sampling:



$$\frac{w_{l \rightarrow m}}{w_{m \rightarrow l}} = \frac{P_m^{eq}}{P_l^{eq}} = e^{-\beta(E_m - E_l)} \equiv e^{-\beta \Delta E}$$

Uphill if $ran < \exp(-\Delta E/k_B T)$
Downhill always accepted

N_0 =transient steps
 N_{max} =simulation steps

$$\langle A \rangle = \frac{1}{n_{max} - n_0} \sum_{n > n_0} A_n$$

Faster to average over some of these states

Recapitulation 8/9:

$$\mathcal{H} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i \quad m = \frac{1}{N} \sum_{j=1}^N \langle s_j \rangle \quad s_i = m + (s_i - m)$$

$$\begin{aligned} s_i s_j &= (m + (s_i - m))(m + (s_j - m)) \\ &= m^2 + m(s_i - m) + m(s_j - m) + (s_i - m)(s_j - m) \end{aligned}$$

ignore

$$E_{MF} = -J \sum_{\langle ij \rangle} (-m^2 + m(s_i + s_j)) - h \sum_i s_i$$

$$-J \sum_{\langle ij \rangle} (-m^2) = \frac{JNz}{2} m^2$$

$$\sum_{\langle ij \rangle} (s_i + s_j) = z \sum_j s_j$$

Where z=4 for square lattice

$$E_{MF} = \frac{JNz}{2} m^2 - (Jzm + h) \sum_{j=1}^N s_j$$

$$Z_{MF} = \sum_{\{s_j\}} \exp(-\beta E_{MF})$$

$$= \exp\left(-\beta \frac{NJzm^2}{2}\right) \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \prod_{j=1}^N \exp(\beta(Jzm + h)s_j)$$

Recapitulation 8/9:

Mean Field Solution – Free energy landscape



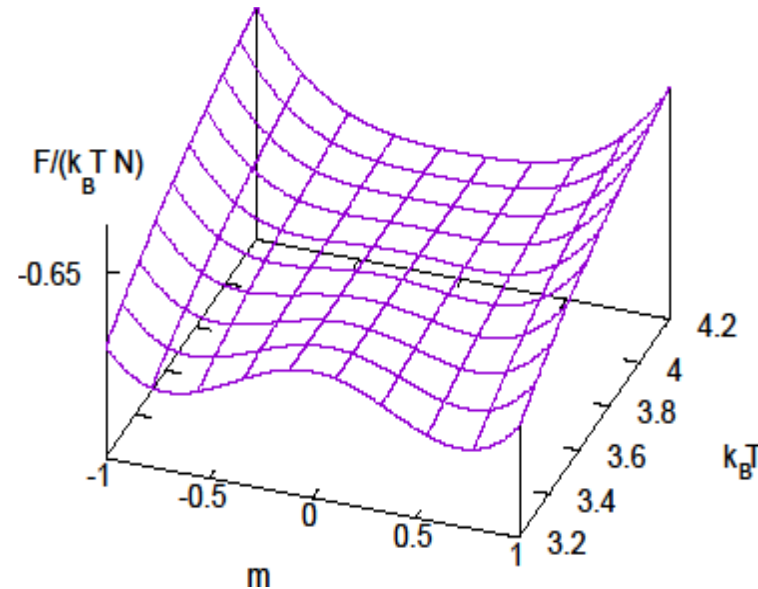
$$\begin{aligned}
 Z_{MF} &= \sum_{\{s_j\}} \exp(-\beta E_{MF}) \\
 &= \exp\left(-\beta \frac{NJzm^2}{2}\right) \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \prod_{j=1}^N \exp(\beta(Jzm + h)s_j) \\
 &= \exp\left(-\beta \frac{NJzm^2}{2}\right) \left[\sum_{s_1=\pm 1} \exp(\beta(Jzm + h)s_1) \right] \cdots \left[\sum_{s_N=\pm 1} \exp(\beta(Jzm + h)s_N) \right] \\
 &= \exp\left(-\beta \frac{NJzm^2}{2}\right) [e^{\beta(Jzm+h)} + e^{-\beta(Jzm+h)}] \cdots [e^{\beta(Jzm+h)} + e^{-\beta(Jzm+h)}] \\
 &= \exp\left(-\beta \frac{NJzm^2}{2}\right) [2 \cosh(Jzm\beta + h\beta)]^N.
 \end{aligned}$$



$$\begin{aligned}
 \frac{f_{MF}}{zJ} &= \frac{F}{zJ \cdot N} = \frac{m^2}{2} - \frac{T}{T_c} \ln 2 - \frac{T}{T_c} \ln \left(\cosh \left(\frac{T_c(m + h')}{T} \right) \right) \\
 &= \frac{m^2}{2} - \theta \ln 2 - \theta \ln \cosh \left(\frac{m + h'}{\theta} \right)
 \end{aligned}$$

$$\theta = T/T_c \quad h' \equiv h/zJ$$

$$\frac{f_{MF}}{zJ} = \frac{m^2}{2} - \theta \ln 2 - \theta \ln \cosh \left(\frac{m + h'}{\theta} \right)$$



$h=0$ case

For $h'=0$ this is symmetric around $m=0$

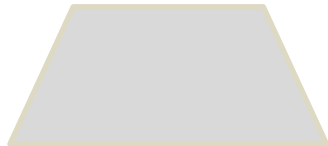
Details, expansion of Free energy landscape:

Using $\text{Cosh}(x) = 1 + x^2/2! + x^4/4! + \dots$ for small $x = m/\theta$

$$\frac{f_{MF}}{zJ} = \frac{m^2}{2} - \theta \ln 2 - \theta \ln \left[1 + \frac{1}{2} \frac{m^2}{\theta^2} + \frac{1}{24} \frac{m^4}{\theta^4} + \mathcal{O}\left(\frac{m^6}{\theta^6}\right) \right]$$

Using $\ln(1+x) = x - x^2/2 + \dots$

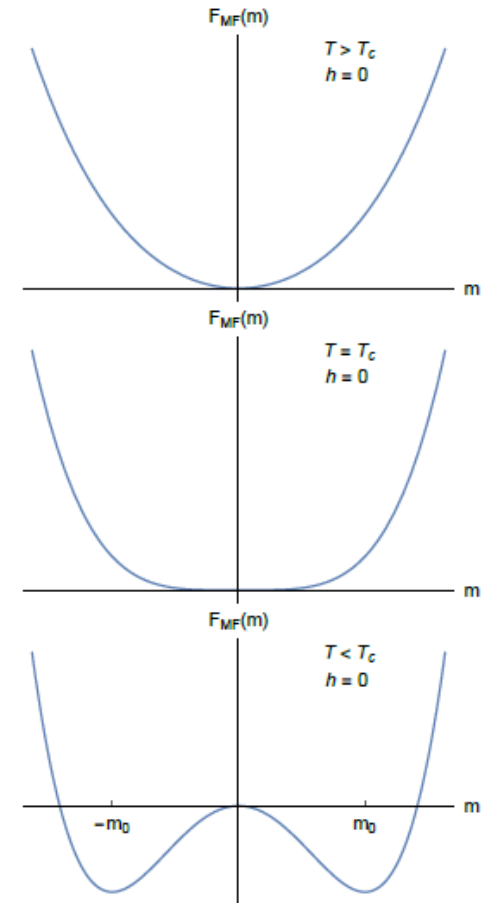
$$\begin{aligned} &= \frac{m^2}{2} - \theta \ln 2 - \theta \left[\frac{1}{2} \frac{m^2}{\theta^2} - \frac{1}{12} \frac{m^4}{\theta^4} + \mathcal{O}\left(\frac{m^6}{\theta^6}\right) \right] \\ &= \frac{m^2}{2} \left(1 - \frac{1}{\theta} \right) - \theta \ln 2 + \frac{m^4}{12\theta^3} + \mathcal{O}\left(\frac{m^6}{\theta^6}\right) \end{aligned}$$



$$1 - 1/\theta = (T - T_c)/T$$

Is positive for $T > T_c$

And Negative for $T < T_c$



Critical point:

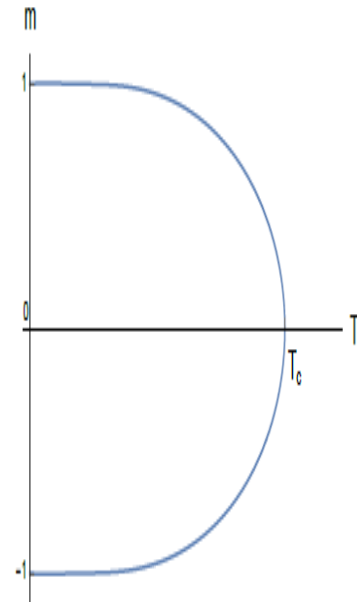
$df_{MF}/dm=0$ and $d^2f_{MF}/d^2m = (1-1/\theta)=(T_c-T)/T$ shifts sign

$$\frac{1}{zJ} \frac{\partial f_{MF}}{\partial m} = m \left(1 - \frac{1}{\theta} \right) + \frac{1}{3} \frac{m^3}{\theta^3} = 0$$

$$m^2 = 3(1 - \theta)\theta^2 = 3(-t)\theta^2$$

$$m(t) = \pm \sqrt{3} \theta |t|^{1/2} = \pm \sqrt{3 \cdot \frac{T^2}{T_c^2} \cdot \frac{T_c - T}{T_c}} \text{ for } T < T_c$$

$$m(t) = 0 \text{ for } T > T_c$$



An external magnetic field h would just add $h \times m$ to

$$\frac{f_{MF}}{zJ} = \frac{m^2}{2} - \theta \ln 2 - \theta \ln \left[1 + \frac{1}{2} \frac{m^2}{\theta^2} + \frac{1}{24} \frac{m^4}{\theta^4} + \mathcal{O} \left(\frac{m^6}{\theta^6} \right) \right]$$

And thus $df_{MF}/dm=0 \rightarrow h=m^3$ for $T=T_c$ ($\theta=0$)
or $\delta=3$ (scaling of h with m along isotherm $T=T_c$)

MF exponents compared:

MF Ising:

$\alpha=0$ $\beta=1/2$ $\gamma=1$ $\delta=3$ No space

Universality class	symmetry	α	β	γ	δ	ν	η
2D Ising	2-component scalar	0	1/8	7/4	15	1	1/4
3D Ising	2-component scalar	0.10	0.33	1.24	4.8	0.63	0.04
3D XY	2-dimensional vector	0.01	0.34	1.3	4.8	0.66	0.04
3D Heisenberg	3-dimensional vector	-0.12	0.36	1.39	4.8	0.71	0.04
mean field		0	1/2	1	3	1/2	0
2D Potts, $q = 3$	q -component scalar	1/3	1/9	13/9	14	5/6	4/15

Table 2.1: Critical exponents for several models.

$C_H \sim |t|^{-\alpha}$ zero-field specific heat

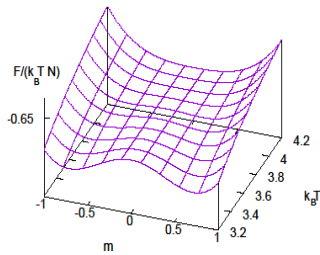
$M \sim (-t)^\beta$ zero-field magnetization

$\chi_T \sim |t|^{-\gamma}$ zero-field isothermal susceptibility

$H \sim |M|^\delta \text{sgn}(M)$ critical isotherm ($t = 0$)

$\xi \sim |t|^{-\nu}$ correlation length

$G(\mathbf{r}) \sim \frac{1}{r^{d-2+\eta}}$ pair-correlation function at T_c .



Landau Theory

$$\left[\begin{aligned} &= \frac{m^2}{2} - \theta \ln 2 - \theta \left[\frac{1}{2} \frac{m^2}{\theta^2} - \frac{1}{12} \frac{m^4}{\theta^4} + \mathcal{O}\left(\frac{m^6}{\theta^6}\right) \right] \\ &= \frac{m^2}{2} \left(1 - \frac{1}{\theta}\right) - \theta \ln 2 + \frac{m^4}{12\theta^3} + \mathcal{O}\left(\frac{m^6}{\theta^6}\right) \end{aligned} \right]$$

- Free energy as a function of the order parameter, $-m/+m$ symmetric:

$$\mathcal{F}(m) = \mathcal{F}_0 + a_2 m^2 + a_4 m^4 + \text{higher order terms}$$

- $a_2 > 0$: $\mathcal{F}(m)$ has only one minimum, namely at $m = 0$,
- $a_2 < 0$: $\mathcal{F}(m)$ has two additional minima,
- $a_2 = 0$: This is the transition between the case of a single minimum and three minima.

Assume a_2 is linear in $t = (T_c - T)/T_c$: $a_2 = a t$:

$$\frac{d\mathcal{F}(m)}{dm} = 0 = 2\tilde{a}_2 t m + 4a_4 m^3 = m(2\tilde{a}_2 t + 4a_4 m^2), = 0$$



$$m = 0$$

$$m = \pm \sqrt{t\tilde{a}_2/2a_4}$$

Hence
 $\beta = 1/2$

Question: Why is there no m^3 term? Why is there no m^6 term?

Question: How would an external field h modify $F(m)$?

Landau with external field

$$\mathcal{F} = \mathcal{F}_0 - hm + \bar{a}_2 t m^2 + a_4 m^4$$

$$\frac{d\mathcal{F}}{dm} = -h + 2\bar{a}_2 t m + 4a_4 m^3 = 0$$

Isoterm, $t=0$

$$h \sim m^3$$

$\delta=3$

$$\chi_T(T, h) \equiv \left(\frac{\partial m}{\partial h} \right)_T$$

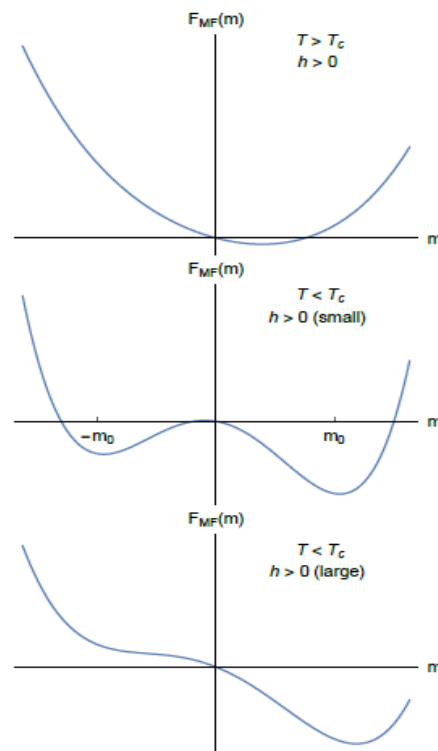
$$= d/dh [h/t]$$

$$= 1/t$$

Thus: $\gamma=1$

when $t>0$,

i.e $T < T_c$



For Ising model this needed
A 2-step expansion
of $\tanh(\beta^*(h+zJm)) \dots$
First around $h=0$ then
around $zJm=0$.

MF exponents (Landau=MF Ising):

Landau Theory:

$\alpha=0$ $\beta=1/2$ $\gamma=1$ $\delta=3$

Universality class	symmetry	α	β	γ	δ	ν	η
2D Ising	2-component scalar	0	1/8	7/4	15	1	1/4
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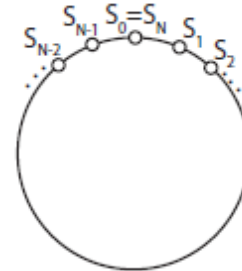
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1-d Ising model

$$\mathcal{H}_N = -J \sum_{i=0}^{N-1} s_i s_{i+1} - h \sum_{i=0}^{N-1} s_i$$



Periodic boundary

$$s_N = s_0.$$

The partition function for the N sites is

$$Z_N = \sum_{s_0=\pm 1, s_1=\pm 1, \dots, s_{N-1}=\pm 1} \exp(\beta J(s_0 s_1 + s_1 s_2 + \dots + s_{N-1} s_0) + \beta h(s_0 + s_1 + \dots + s_{N-1}))$$

$$Z_N = \sum_{s_0=\pm 1, s_1=\pm 1, \dots, s_{N-1}=\pm 1} \underbrace{e^{\beta J s_0 s_1 + \beta h \frac{s_0 + s_1}{2}}}_{T_{0,1}} \underbrace{e^{\beta J s_1 s_2 + \beta h \frac{s_1 + s_2}{2}}}_{T_{1,2}} \dots$$

$$\cdot \underbrace{e^{\beta J s_{N-1} s_0 + \beta h \frac{s_{N-1} + s_0}{2}}}_{T_{N-1,0}} \cdot$$

Each factor can take one of 4 different values,.. Can be solved....
But a little too technical here (see questions for Helix-Coil transition)

Spatial correlations in 1d:

$$\begin{aligned}
 \langle s_i s_{i+l} \rangle &\equiv \frac{1}{Z} \sum_{s_1=\pm 1, s_2=\pm 2, \dots, s_N=\pm 1} s_i s_{i+l} e^{\beta J \sum_k s_k s_{k+1}} \\
 &= \frac{1}{Z} \lim_{J_k \rightarrow J, \text{ for } k \in \{1, \dots, N\}} \sum_{\text{all states}} \underbrace{s_i s_{i+1}}_1 \underbrace{s_{i+1} s_{i+2}}_1 \underbrace{s_{i+2} s_{i+3}}_1 \dots \\
 &\quad \cdot \underbrace{s_{i+l-1} s_{i+l}}_1 e^{\beta \sum_k J_k s_k s_{k+1}} \\
 &= \frac{1}{Z} \lim_{J_k \rightarrow J, \text{ for } k \in \{1, \dots, N\}} \sum \frac{1}{\beta^l} \frac{\partial^l Z}{\partial J_i \partial J_{i+1} \partial J_{i+2} \dots \partial J_{i+l-1}} \\
 &= \frac{[\sinh(\beta J)]^l [\cosh(\beta J)]^{N-l}}{[\cosh(\beta J)]^N} \\
 &= [\tanh(\beta J)]^l \\
 &= \exp(l \ln(\tanh(\beta J))) \\
 &= \exp \left(-l \ln \left(\underbrace{\frac{1}{\tanh(\beta J)}}_{1/\ell} \right) \right),
 \end{aligned}$$

Trick: inserting factors =1 that make telescope product,

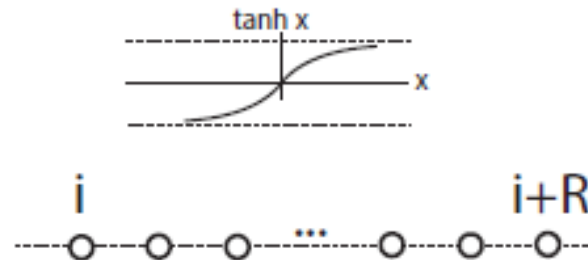


differentiate

$$Z = \sum_{\text{all states}} \exp(\beta \sum_k J_k s_k s_{k+1})$$

with respect to $J_i, J_{i+1}, \dots$ to J_{i+l-1}

Each neighbor spin pair gives one factor $\cosh()$ for whether the two spins are aligned or not
 $\rightarrow$ differentiation gives $\sinh()$.



This is the probability that one coherent region has length ℓ
 Regions always have finite length!

Intuitively:

Spin aligned, or not:

For each step

$$\Gamma(1) = \langle s_i s_{i+1} \rangle = \frac{\sinh \beta J}{\cosh \beta J} = \tanh \beta J$$

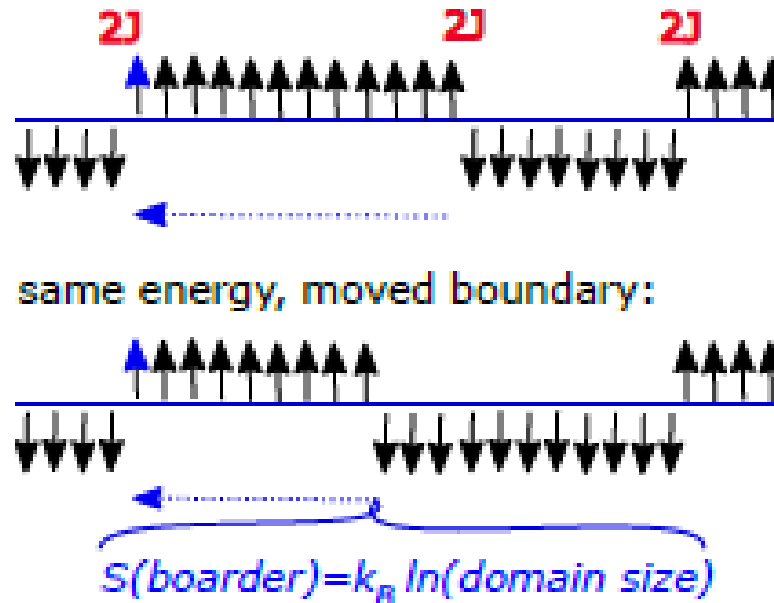
Normalized for proper weight

Can only be correlated
Involving all steps in between:

$$\Gamma(l) = \langle s_i s_{i+l} \rangle = \langle s_i s_{i+1} s_{i+1} s_{i+2} s_{i+2} \dots s_{i+l} \rangle = \tanh^l \beta J$$

This reflects the fact that it is entropically impossible to have
Infinitely large segments of spins in the same direction...
Each boundary will give entropy $S = \ln(\ell)$ and when
 $k_B T \ln(\ell) > J$ then it is favorable to break up the system on scale ℓ

... fundamentally different than 2 dimensions



Energy of border = $2J$

Entropy of border = $k_b \ln(\text{domain size})$

Balance when: $2J = k_b T \ln(\text{domain size}) \rightarrow \text{domain size} = \exp(+2\beta J)$

Compared to $\rightarrow$

$$= \exp(l \ln(\tanh(\beta J)))$$

$$= \exp \left(-l \ln \left(\underbrace{\frac{1}{\tanh(\beta J)}}_{1/\xi} \right) \right),$$

$$\begin{aligned} \xi &= -1 / \ln[(\exp(\beta J) - \exp(-\beta J)) / (\exp(\beta J) + \exp(-\beta J))] \\ &= -1 / \ln[1 - 2\exp(-\beta J) / (\exp(\beta J) + \exp(-\beta J))] \\ &\sim 1 / [2\exp(-\beta J) / (\exp(\beta J) + \exp(-\beta J))] \\ &\sim \exp(+2\beta J) / 2 \text{ diverges with } T \rightarrow 0 \end{aligned}$$

In more detail,

Energy of border = $2J$

Entropy of borders = $k_b \ln(\text{domain size})$

$L = \text{system size}$

$X = \text{number of borders}$

Minimize $F = 2Jx - x k_b T \ln(L/x)$ with respect to x :

$$dF/dx = 0 \rightarrow 2J - k_b T \ln(L/x) + x k_b T/x = 0 \rightarrow$$

$$L/x = \exp(1 + 2J/k_b T)$$

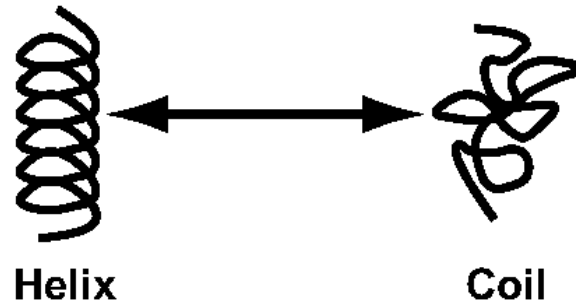
Thus domain size $\sim \exp(+2\beta J)$

Notice that x borders among L sites can be placed at

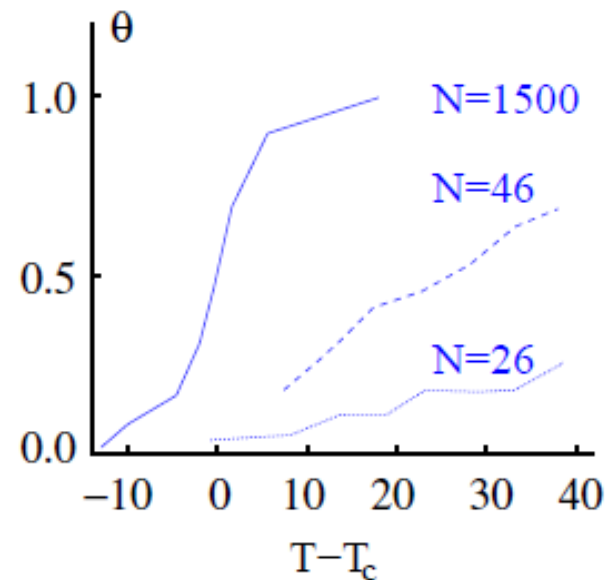
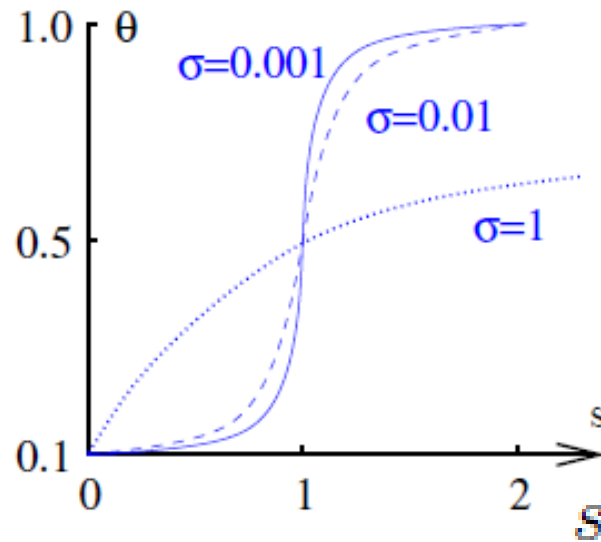
$L!/(x!(L-x)!)$ sites, $\rightarrow$

$$\begin{aligned} \text{Full entropy} &= \ln(L!) - \ln(x!) - \ln((L-x)!) = L \ln L - Lx \ln(x) + (L-x) \ln(L-x) \\ &= x \ln(L) - x \ln(x) + x \ln(1-x/L) - L \ln(1-x/L) \\ &= x \ln(L/x) + (x-L) \ln(1-x/L) \\ &\sim x \ln(L/x) + (1-x/L)x/L \end{aligned}$$

Helix coil transition:



Helix-coil transition with system size



$$\Theta = \frac{s}{2\lambda_{max}} \left(1 + \frac{s + 2\sigma - 1}{\sqrt{(1-s)^2 + 4\sigma s}} \right)$$



Right: The used data is the temperature-induced coil-to-helix transition of the polypeptide poly- γ -benzyl-L-glutamate in mixed dichloroacetic acid-ethylene dichloride solvent (Zimm, Doty and Iso (1959)). Usually the helix fraction would decrease with T , but for this polymer the helix stabilizing forces increases more with temperature than the contribution from the entropy.

Transition, but not a phase transition.... Small $\sigma \rightarrow 0$ with $T \rightarrow 0$ (cost of boundary high at $T=0$)

EXTRA

1d Ising model $\leftrightarrow$ Helix-Coil transitions

The helix-coil transition can be parametrized in terms of two quantities (Zimm (1959), Sheraga (1973)): an entropy cost term σ that counts the loss in conformational entropy to form the first bond, and an energy gain term that counts the energy for each subsequent addition of a bond. In more detail, let us normalize the statistical weights relative to the coil (c) state, i.e. we assign to it a free energy $F_c = 0$ and therefore a statistical weight $\exp(-F_c/k_B T) = 1$. To initiate a helix then costs a free energy $F_{ch} = F_{hh} - T S_{ch}$ with a corresponding statistical weight $\exp(-F_{ch}/k_B T) = \sigma s$; to continue a helix costs a free energy F_{hh} , with a statistical weight $\exp(-F_{hh}/k_B T) = s$. From the above, $\sigma < 1$ always, while s can be < 1 (then there is no transition to a helix state), or > 1 . Thus:

σs = statistical weight to initiate helix in coil region

s = statistical weight to continue helix

$$Z(\text{cccccccchhhhhhhhhhhccccchhhhhhh}) = \sigma^j s^k = \sigma^2 s^{17}$$



1/Cost of boundary

Calculate:

$$Z = \sum_{\{\text{sequences } s\}} \exp\left(-\frac{F\{s\}}{k_B T}\right)$$

$$c \rightarrow h: \sigma s$$

$$c \rightarrow c: 1$$

$$h \rightarrow h: s$$

$$h \rightarrow c: 1$$

$$Z(N) = Z_c(N) + Z_h(N)$$

$$Z_c(N+1) = Z_c(N) + Z_h(N)$$

$$Z_h(N+1) = s\sigma Z_c(N) + s Z_h(N)$$

$$\begin{pmatrix} Z_c(N+1) \\ Z_h(N+1) \end{pmatrix} = \mathbf{M} \begin{pmatrix} Z_c(N) \\ Z_h(N) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \sigma s & s \end{pmatrix} \begin{pmatrix} Z_c(N) \\ Z_h(N) \end{pmatrix}$$

Transfer matrix for Helix-coil:

Assume start in coil state (at index=0), then:

$c \rightarrow h: \sigma s$
 $c \rightarrow c: 1$
 $h \rightarrow h: s$
 $h \rightarrow c: 1$

$$Z_c(1) = 1, Z_h(1) = \sigma s, \quad \text{or} \quad \begin{pmatrix} 1 \\ \sigma s \end{pmatrix} = \mathbf{M} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$F = -k_B T \ln(Z) = -k_B T N \ln(\lambda_{max}) = -k_B T N \ln\left(\frac{1}{2}(1+s+\sqrt{(1-s)^2+4\sigma s})\right)$$

$$\begin{pmatrix} Z_c(N) \\ Z_h(N) \end{pmatrix} = \mathbf{M}^N \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda \mathbf{1}) = 0$$

$$(1 - \lambda) \cdot (s - \lambda) = \sigma s$$

$$\lambda = \frac{1}{2} \left(s + 1 \pm \sqrt{(1-s)^2 + 4\sigma s} \right)$$

$$Z(N) = Z_c(N) + Z_h(N) \sim \lambda_{max}^N$$

$$F = -k_B T \ln(Z) = -k_B T N \ln(\lambda_{max}) = -k_B T N \ln\left(\frac{1}{2}(1+s+\sqrt{(1-s)^2+4\sigma s})\right)$$

Helix-coil transition:

The point is that the partition sum can also be expressed as

$$Z = \sum \Omega_{jk} \sigma^j s^k$$

Now the number of monomers in helix state (s)

$$\frac{d \ln(Z)}{d \ln(s)} = \frac{s}{Z} \frac{dZ}{ds} = \frac{\sum \Omega_{jk} k \sigma^j s^k}{\sum \Omega_{jk} \sigma^j s^k} = \langle k \rangle$$

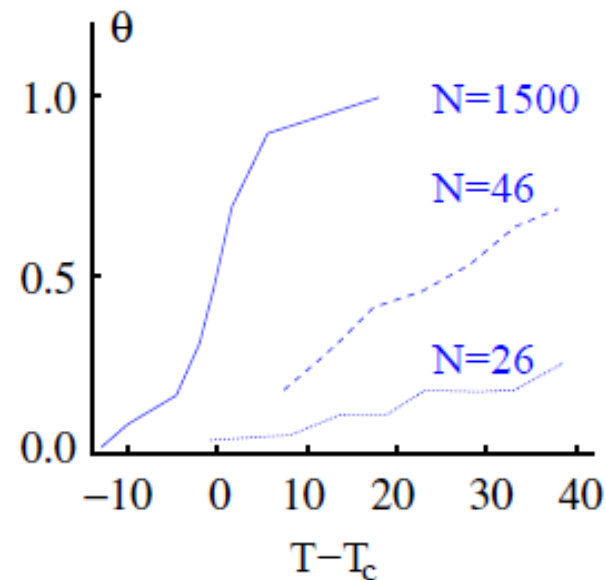
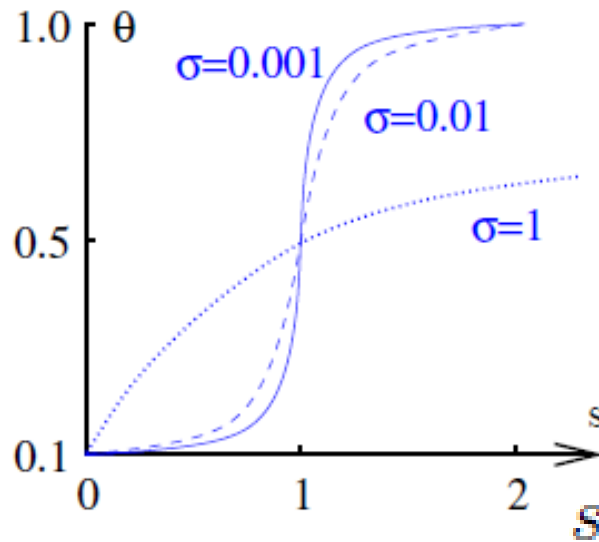
$$\Theta = \frac{1}{N} \frac{d \ln(Z)}{d \ln(s)} = \frac{s}{2\lambda_{max}} \left(1 + \frac{s + 2\sigma - 1}{\sqrt{(1-s)^2 + 4\sigma s}} \right)$$

number of helix initiations:

$$\langle j \rangle = \frac{1}{N} \frac{d \ln(Z)}{d \ln(\sigma)} = \frac{s\sigma}{\lambda_{max} \sqrt{(1-s)^2 + 4\sigma s}}$$

EXTRA

Helix-coil transition with system size



$$\Theta = \frac{s}{2\lambda_{max}} \left(1 + \frac{s + 2\sigma - 1}{\sqrt{(1-s)^2 + 4\sigma s}} \right)$$



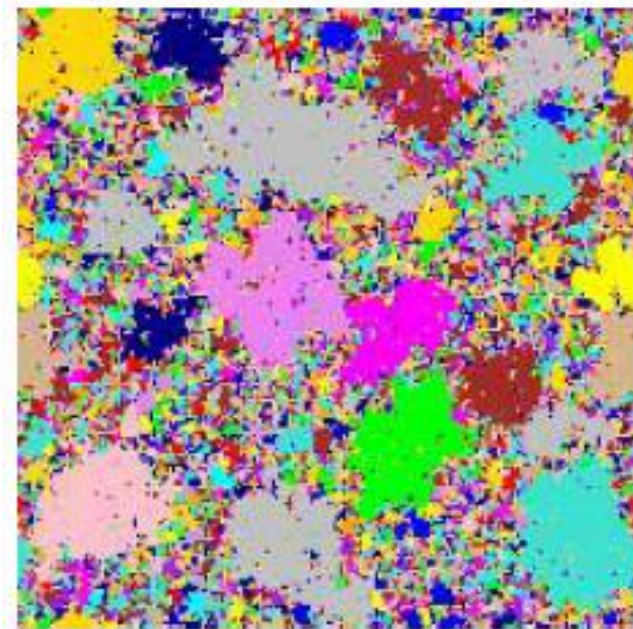
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Variant model with 1st order transition: The Potts model

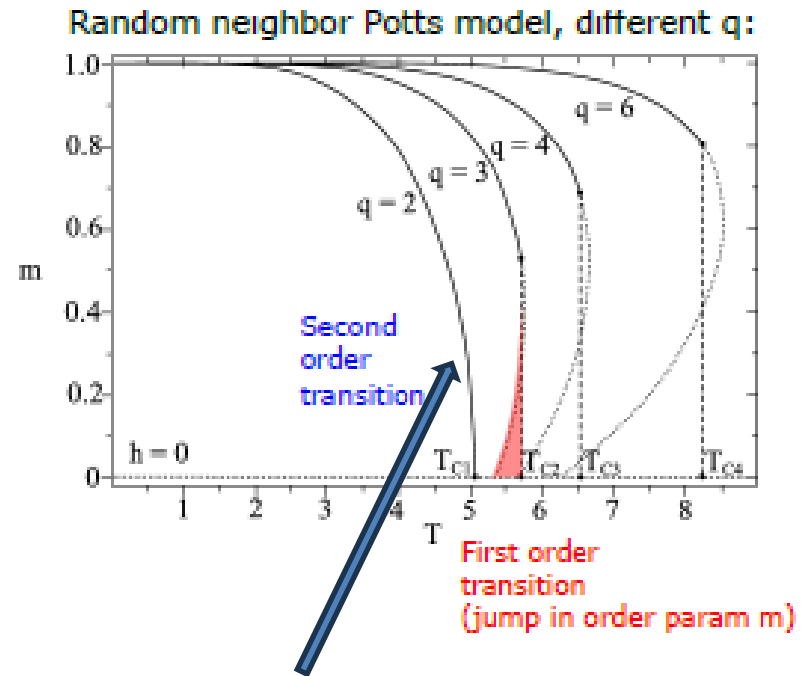
$$H = - \sum_{\langle i,j \rangle} \delta(s_i - s_j)$$

Where s_i takes one of q values and sum over the nearest neighbors



$q=16$

Figure 1.9: 16 state Potts model



$q=2$ is Ising model

Potts model mean field

$$H = \frac{1}{N} \sum_{v>u} \delta_{s(v)-s(u)} \quad i=0,1,\dots,q-1 \text{ states:} \quad \sum_i x_i = 1$$

Energy and Entropy and Free energy:

$$\frac{E}{N} = -(1/2) \sum_i x_i^2$$

$$\frac{S}{k_B N} = - \sum_i x_i \ln(x_i)$$

Entropy is biggest if all is equally represented

$$\beta \frac{F}{N} = \sum_i \left(x_i \ln(x_i) - \frac{1}{2} \beta x_i^2 \right)$$

(remember $F=E-TS$)

Order parameter s:

$$x_0 = \frac{1}{q} (1 + (q-1)s)$$

$$x_i = \frac{1}{q} (1 - s) \quad \text{for all } i = 1, 2, \dots, q-1$$

With $s=0$ corresponding to all $x_i=1/q$,
And $s=1$ corresponding to $x_0=1$, all other $=0$

Potts model mean field; Free energy

$$\begin{aligned}
 \beta \left(\frac{F(s)}{N} - \frac{F(s=0)}{N} \right) = & + \frac{1 + (q-1)s}{q} \cdot (\ln(1 + (q-1)s) - \ln(q)) \\
 & + (q-1) \frac{1-s}{q} \cdot (\ln(1-s) - \ln(q)) \\
 & - \frac{1}{2} \beta \left(\frac{(1 + (q-1)s)^2}{q} \right) - \frac{1}{2} \beta \cdot (q-1) \left(\frac{(1-s)^2}{q} \right) \\
 & + \frac{1}{2} \beta q \cdot \left(\frac{1}{q} \right)^2 \\
 = & + \frac{1 + (q-1)s}{q} \cdot \ln(1 + (q-1)s) \\
 & + (q-1) \frac{1-s}{q} \cdot \ln(1-s) - q \ln(q) \\
 & - \frac{1}{2} \beta \cdot \left(\frac{q-1}{q} s^2 \right)
 \end{aligned}$$

$\downarrow S(s=0)$
 $\left. \begin{array}{l} \leftarrow X_0 \\ \leftarrow X_{\dots} \end{array} \right\}$ Entropy part
 Energy of s
 Energy of s=0

Possible
stable states



$$\begin{aligned}
 \frac{dF/N}{ds} &= \frac{(q-1)}{q} \ln(1 + (q-1)s) + \frac{(1 + (q-1)s)(q-1)}{q(1 + (q-1)s)} \\
 &- \frac{q-1}{q} \ln(1-s) - \frac{(q-1)(1-s)}{q(1-s)} - \beta \frac{q-1}{q} s \\
 &= \frac{(q-1)}{q} (\ln(1 + (q-1)s) - \ln(1-s) - \beta s) = 0 \\
 \Rightarrow \beta s &= \ln\left(\frac{1 + (q-1)s}{1-s}\right)
 \end{aligned}$$

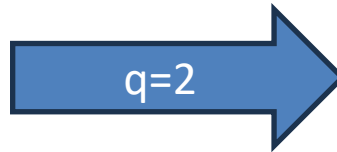
Potts model, stable states:

Possible
stable states



$$\begin{aligned}\frac{dF/N}{ds} &= \frac{(q-1)}{q} \ln(1 + (q-1)s) + \frac{(1 + (q-1)s)(q-1)}{q(1 + (q-1)s)} \\ &\quad - \frac{q-1}{q} \ln(1-s) - \frac{(q-1)(1-s)}{q(1-s)} - \beta \frac{q-1}{q} s \\ &= \frac{(q-1)}{q} (\ln(1 + (q-1)s) - \ln(1-s) - \beta s) = 0 \\ \Rightarrow \beta s &= \ln\left(\frac{1 + (q-1)s}{1-s}\right)\end{aligned}$$

$$\beta s = \ln\left(\frac{1 + (q-1)s}{1-s}\right)$$



$$s = \tanh(\beta \cdot s/2)$$

$$\beta \left(\frac{F(s)}{N} - \frac{F(s=0)}{N} \right) =$$

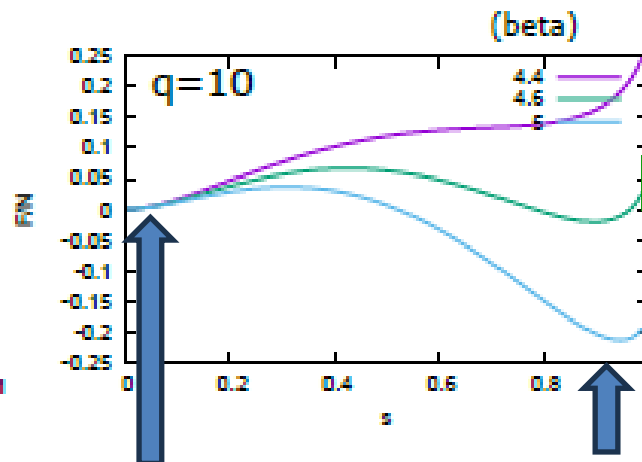
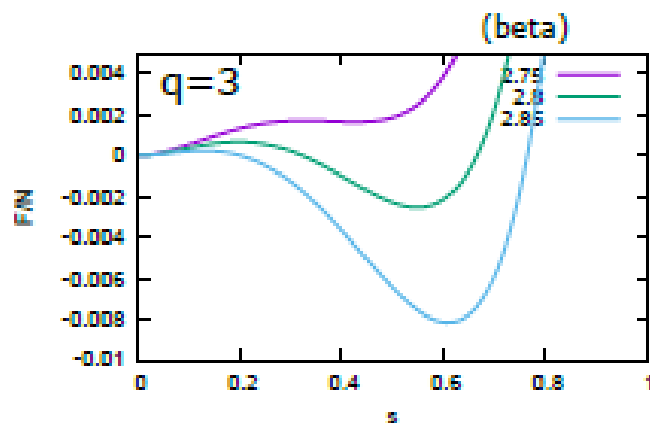
$$= + \frac{1 + (q-1)s}{q} \cdot \ln(1 + (q-1)s) + (q-1) \frac{1-s}{q} \cdot \ln(1-s) - q \ln(q) - \frac{1}{2} \beta \cdot \left(\frac{q-1}{q} s^2 \right)$$



$$= \frac{q-1}{2q} \left(q - \frac{\beta}{2} \right) s^2 - \frac{1}{6} (q-1)(q-2) s^3 + \dots$$

$q > 2$, First order transition
(extremum at $s \sim 0$
and at higher s since $s < 1$)

For larger q , at some T there is more than one solution for s :

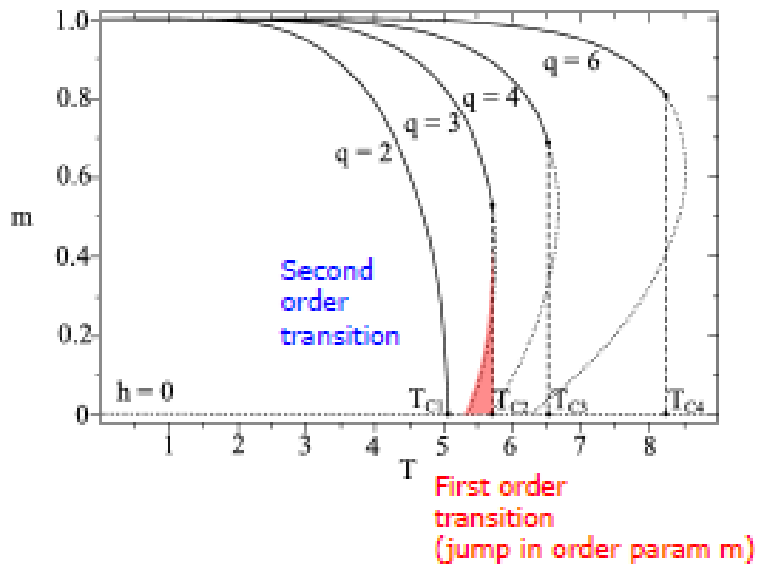


“democracy”
(or anarchy)

Dictatorship

Some T values
Where there
are two
**Qualitatively
Different solutions.**
With different order!
Hysteresis $\rightarrow$
First order

Random neighbor Potts model, different q :



Finite jump
in order parameter,
and in Energy and S
(but same value of F)

